

Prediction Targets for Cross-Sectional Bond Selection

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Abstract

Machine-learning models are usually trained to forecast returns but used to sort securities into long-short portfolios. I show that this mismatch matters in corporate bonds. A return forecast partly predicts the market-wide movement common to all bonds. This movement does not help to pick one bond over another, and fitting it distorts the ranking. Holding the model and data fixed, I replace the return target with the probability that a bond beats the cross-sectional median. The rank information coefficient rises from 0.137 to 0.204, and the Sharpe ratio of the long-short portfolio from 1.17 to 2.21. The gain comes mainly from lower risk rather than higher average returns, as the theory predicts. The long-short portfolio formed on the probability forecast has half the bond-market beta of the one formed on the return forecast, and when both portfolios are scaled to the same level of risk, the probability-based portfolio earns 1.40% per month against 0.65%. Forecasting the probability of a positive return does not improve the ranking, so the gain comes from making the target relative, not from the switch to classification. The result holds for other machine-learning models, including gradient-boosted trees and neural networks, and the advantage becomes larger when bonds are compared only with bonds of the same rating or credit spread.

Keywords: corporate bonds; machine learning; probability forecasts; return prediction; cross-sectional selection.

JEL: G11, G12, G17, C53.

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1 Introduction

Machine learning has changed the empirical technology used to forecast asset returns. Flexible models accommodate nonlinearities, interactions, and high-dimensional predictor sets that are difficult to represent with conventional linear specifications. In both equities and corporate bonds, this literature has produced large improvements in statistical fit and portfolio performance (Gu et al., 2020; Bali et al., 2022; Kelly et al., 2023; Bell et al., 2024). Yet the standard empirical design embeds a choice that has received much less attention than model architecture: *what should the model be asked to predict?*

The conventional answer is the next-period return. A model is trained under squared-error loss to approximate the conditional mean, $\mathbb{E}_t[R_{i,t+1}]$. The fitted values are then used to rank securities, and the economic value of the model is assessed from rank correlations or long-short portfolios. This procedure is natural if accurate point forecasts are themselves the object of interest. It is less obvious when the final decision is purely cross-sectional. A ranking investor does not trade the level of next month's bond-market return. The investor trades differences between securities. Any component common to all bonds in a month affects return-forecast errors but cannot change the ordering of securities or the return on a dollar-neutral long-short portfolio.

Corporate bonds provide a useful setting in which to study this distinction. Monthly bond returns contain substantial common variation associated with Treasury rates, aggregate credit conditions, and market-wide risk. In my sample, month fixed effects explain 42% of the variance of monthly excess returns, and in the evaluation period the monthly cross-sectional mean accounts for 47.5% of the total sum of squared excess returns. A return-forecasting model can therefore improve squared-error fit by learning movements in the common level of bond returns. That prediction may be valuable for market timing, but it is not useful for choosing one bond over another. The statistical objective used in training and the economic objective used in evaluation can consequently reward different dimensions of predictability.

This paper studies the forecasting target as an economic design choice. The empirical strategy is a controlled target experiment. I hold fixed the information set, the learner, the training and validation windows, the real-time estimation protocol, and the out-of-sample evaluation. I vary only the outcome the model is asked to learn. The benchmark target is the next-month excess return, estimated under squared error. I compare it with three binary events estimated under binary cross-entropy: a positive excess return, outperformance of the contemporaneous cross-sectional mean, and outperformance of the contemporaneous cross-sectional median. The design separates the effect of the target from the effect of the machine-learning architecture.

The most informative comparison holds the loss function fixed. Both the probability of a positive excess return and the probability of outperforming the cross-sectional mean are estimated with binary cross-entropy. They differ only in the benchmark that defines success. The first is an absolute event, $R_{i,t+1} > 0$. The second is a relative event, $R_{i,t+1} > \bar{R}_{t+1}$. If classification or binary cross-entropy is responsible for superior ranking, both probability forecasts should improve on the return model. If the relevant distinction is whether the target is aligned with cross-sectional selection, only the relative event should improve ranking.

Section 3 formalizes this distinction. I decompose each return into a common component and a relative component and state all moments under the pooled distribution that pooled estimation uses. The return of a dollar-neutral portfolio depends only on the relative component, and the out-of-sample R^2 of a return forecast decomposes exactly into a time-series part and a cross-sectional part. The central result shows that the slope of a pooled return projection equals the slope of a projection on relative returns plus a between-month contamination term. The term is nonzero whenever the average levels of the characteristics vary over time and forecast the common component, which is the normal state of affairs in corporate bonds, where average spreads and yields move with the credit cycle. The return target then ranks bonds as well as the best linear score only in a knife-edge case. Relative targets whose cross-sectional mean is constant have no between-month term at all, whereas the absolute event inherits the contamination because the fraction of bonds with positive returns rises with the common component. A second result shows that, under a factor structure with heterogeneous exposures, a forecast of expected relative returns rewards exposure linearly while the probability of beating the cross-sectional benchmark rewards it at most by the Sharpe ratio of the common factor.

The results favor the relative interpretation. In annual expanding-window tests from January 2012 through August 2020, an explainable boosting machine (EBM) trained on next-month returns produces an average monthly Spearman rank information coefficient (IC) of 0.137. Forecasting the probability that a bond outperforms the contemporaneous cross-sectional median raises the IC to 0.204. The monthly difference of 0.067 has a Newey–West t -statistic of 4.99, and the corresponding difference in a fixed chronological split is 0.068 ($t = 6.29$). By contrast, the probability that next month’s excess return is positive produces an IC of 0.135, essentially identical to the return model. Moving the binary threshold from zero to the cross-sectional mean raises the IC by 0.060 ($t = 2.91$) while preserving the same loss. The principal distinction is therefore not regression versus classification. It is absolute versus relative forecasting.

The portfolio comparison shows where the economic gain comes from. The equal-

weighted top-minus-bottom decile portfolio based on the return forecast earns 0.612% per month with a monthly volatility of 1.817%, an annualized Sharpe ratio of 1.17. The portfolio based on the median-outperformance probability earns a similar average spread, 0.727% per month, with much lower volatility, 1.137%, and therefore a Sharpe ratio of 2.21. The difference in mean spreads is not statistically significant, and it is not the relevant comparison. A long-short portfolio can be scaled, so an investor cares about the return per unit of risk. When both strategies are scaled every month to the same ex-ante volatility, the probability strategy earns 1.40% per month against 0.65% for the return strategy, a difference of 0.76% with a t -statistic of 7.99, and the difference in Sharpe ratios is 1.05 ($t = 5.62$). The lower volatility is not a by-product. It is what the framework predicts: a probability forecast rewards exposure to the common factor at most by the factor’s Sharpe ratio, whereas a return forecast rewards it one for one with the factor premium. The relative target should therefore produce a portfolio with less systematic risk at a similar mean, and it does. The return portfolio’s long-short spread has a bond-market beta of 1.00, while that of the probability portfolio has a beta of 0.53.

A comparison with the absolute probability target shows that low volatility alone is not the result. The probability of a positive excess return also produces a low-volatility portfolio, because the event of a positive return is most likely for safe bonds, but it does not rank bonds better than the return model, and its risk-adjusted and alpha advantages are not robust across designs and weighting schemes. The relative target is the only one that combines the two sources of improvement: better ordering, which comes from the relative threshold, and lower systematic risk, which comes from the probability mapping. This is why its gains appear in the rank IC, the Sharpe ratio, and the factor-adjusted alpha at the same time.

Several comparisons show that the result is not a peculiarity of one model. First, the relative-probability advantage survives changes in architecture. In XGBoost, the median target raises the IC from 0.121 to 0.200, a difference of 0.080 ($t = 4.88$). In a three-layer neural network, the IC rises from 0.132 to 0.215, a difference of 0.083 ($t = 5.73$). The neural-network result is informative about the mechanism. Its inputs are ranked within each month, which removes all between-month variation in characteristic levels and therefore shuts down the linear contamination channel. The return target still ranks worse, so part of the advantage comes from the within-month exposure channel.

Second, relative targets generate much greater agreement across model classes. Under the return objective, the average monthly rank correlation between the XGBoost and neural-network forecasts is 0.27, and between EBM and XGBoost 0.68. Under the median-outperformance objective, the corresponding correlations are 0.77 and 0.88. Different

learners disagree substantially about the conditional mean return but converge toward a similar ordering when asked to predict relative performance.

Third, the probability forecast contains information that the return forecast lacks, and not the reverse. In sequential double sorts, sorting on the probability forecast within quintiles of the return forecast earns 0.36% per month ($t = 6.02$) with a Sharpe ratio of 2.72, while sorting on the return forecast within quintiles of the probability forecast earns 0.14% ($t = 1.41$). In Fama–MacBeth regressions that include both forecast ranks, the coefficient on the probability forecast remains significant ($t = 5.13$ for returns and $t = 8.98$ for return ranks), whereas the coefficient on the return forecast becomes insignificant.

Fourth, the relative target performs especially well when broad credit exposures are neutralized. Within monthly rating terciles, the return model’s IC falls to 0.094 while the median-outperformance probability retains an IC of 0.174. Within spread terciles, the corresponding ICs are 0.063 and 0.148, and within rating-by-duration cells 0.098 and 0.194. The IC gap therefore becomes larger, not smaller, when evaluation removes coarse credit and duration differences. The relative target improves security selection within risk groups rather than repackaging a spread or rating sort.

Factor regressions point in the same direction. The EBM return long–short portfolio has a bond-market beta of 1.00 and a monthly three-factor alpha of 0.483%. The median-outperformance portfolio has a bond-market beta of 0.53 and an alpha of 0.671% ($t = 8.31$). The alpha difference is 0.188% per month with a t -statistic of 2.55 in the expanding window and 0.204% with a t -statistic of 5.00 in the fixed split. The mean spreads of the two portfolios are close because a large part of the return portfolio’s spread is compensation for bond-market exposure; once that exposure is priced, the relative target earns significantly more. The long leg of the probability strategy is particularly different: its bond-market beta is 0.17 and its factor R^2 is 0.18, compared with a beta of 0.69 and an R^2 of 0.56 for the return-model long leg. The improvement in risk-adjusted performance reflects the economic content of the ranking rather than a larger loading on systematic credit risk.

Finally, the decomposition of out-of-sample R^2 shows what the return model’s measured accuracy consists of. For the baseline EBM, total out-of-sample R^2 is 0.76%, of which 0.57 percentage points, or 75%, come from prediction of the monthly common component. This does not imply that aggregate bond-market forecasting is unimportant. It shows that headline return-prediction R^2 combines two economically distinct forecasting tasks, only one of which affects a within-month ranking. The advantage of the relative target is concentrated in investment-grade and long-duration bonds, where heterogeneous rate exposure dominates cross-sectional return differences, and it holds in 96 of 100 random assignments of calendar years to estimation and evaluation samples.

These results connect two strands of the recent literature. [Bell et al. \(2024\)](#) show that explainable nonlinear models forecast corporate bond returns and use the EBM structure to characterize what the model learns. I take that return-prediction framework as a benchmark and ask whether the object that maximizes return-prediction fit is also the object that best supports security selection. [He et al. \(2025\)](#) show that probability forecasts contain economically useful information for asset pricing and emphasize the information-ratio interpretation of probability ranking. I use corporate bonds to separate the role of probability from the role of the benchmark defining the event. The finding that $\mathbb{P}(R > 0)$ does not improve ranking while $\mathbb{P}(R > \bar{R})$ does provides a direct identification of the relative component of the probability objective.

The contribution is therefore not a new machine-learning architecture, nor simply an application of classification to corporate bonds. The contribution is to show, theoretically and empirically, that the forecasting target changes the economic content of machine-learning predictions even when the learner and information set are held fixed. In a market with large common return variation, a model optimized for point prediction allocates predictive capacity to variation that the portfolio rule discards, and its cross-sectional ranking is contaminated by the use of characteristic levels to forecast that variation. Relative targets remove this contamination. They improve cross-sectional ranking, reduce systematic exposure, and make predictions more stable across architectures.

The remainder of the paper proceeds as follows. Section 2 relates the analysis to the literature. Section 3 develops the distinction between expected-return prediction and cross-sectional selection. Section 4 describes the sample, the targets and loss functions, the learners, and the real-time evaluation. Section 5 presents the main EBM target comparison, including conditional portfolio sorts that test whether the probability forecast carries information beyond the return forecast. Section 6 extends the comparison to XGBoost and the neural network. Section 7 repeats the conditional sorts for these learners and adds Fama–MacBeth regressions. Section 8 examines the mechanism: what return models forecast, whether the relative target selects bonds or credit exposure, and what the targets learn. Section 9 reports robustness and limitations, and Section 10 concludes. Proofs are in Appendix A.

2 Related Literature

This paper is related to three literatures. The first uses machine learning to estimate expected returns. In equities, [Gu et al. \(2020\)](#) demonstrate that nonlinear models and interactions can improve return prediction relative to linear benchmarks. [Freyberger](#)

et al. (2020) characterize nonlinear characteristic effects nonparametrically, Kozak et al. (2020) study shrinkage in a high-dimensional cross-section, and Kelly et al. (2019) model characteristics as instruments for time-varying factor loadings. A common feature of this literature is that the prediction target is the realized return or a closely related conditional-mean object. The resulting forecasts are frequently evaluated by cross-sectional sorts, even though the loss used for estimation is a point-forecast loss.

Corporate bonds provide a distinct environment for these methods. Bali et al. (2022) show that machine-learning forecasts of bond returns improve on structural and reduced-form benchmarks. Kelly et al. (2023) model the cross-section of expected corporate bond returns using latent factors and characteristics. Bell et al. (2024) show that explainable boosting machines capture economically important nonlinearities and interactions in corporate bond returns while retaining interpretability. Their results make EBM a natural benchmark for the present exercise: rather than asking whether a different architecture predicts returns better, I keep the EBM framework and vary the object it is asked to learn. Work on corporate bond factors, including Bai et al. (2019) and Dickerson et al. (2023), motivates the factor-adjustment and credit-neutral exercises below.

The second literature studies probability forecasts and classification in asset pricing. He et al. (2025) argue that the probability that an asset outperforms a benchmark can contain information distinct from its conditional expected return. Under a location-scale model, an outperformance probability is monotone in a conditional information ratio. This provides a natural explanation for why probability-ranked portfolios can differ from expected-return-ranked portfolios even when both are based on the same predictors. Brownlees and Souza (2025) similarly emphasize classification objectives in portfolio choice. My analysis is complementary but asks a different identification question: when probability forecasting improves cross-sectional performance, how much comes from probability as such and how much from defining the event relative to the contemporaneous cross-section?

That distinction is particularly important in bonds. An event such as $R_{i,t+1} > 0$ remains sensitive to a large common shift in the bond market. If all bonds receive a sufficiently positive aggregate shock, nearly every observation becomes a success. In contrast, the event $R_{i,t+1} > \bar{R}_{t+1}$ is invariant to that common shift. Comparing the two events while holding the loss function fixed isolates an economic feature of the target that a generic regression-versus-classification comparison does not identify. My framework also explains why the threshold matters for estimation, through the between-month contamination of pooled projections.

The third literature studies the relationship between forecasting loss and economic

decisions. Forecasting theory has long emphasized that the optimal forecast depends on the loss under which it is evaluated (Granger, 1969; Christoffersen and Diebold, 1997; Elliott and Timmermann, 2016), and that competing forecasts should be compared with tests that account for their correlation (Diebold and Mariano, 1995). Asset-pricing applications introduce an additional layer because statistical models are often estimated under one loss and consumed by a different decision rule. A cross-sectional portfolio is invariant to transformations that can materially change squared-error loss. The present setting makes this mismatch transparent and measurable. Rather than propose a new loss function, I use economically interpretable target definitions to identify which parts of the return outcome matter for the downstream ranking problem.

The paper also relates to the distinction between market timing and security selection. Predicting the common monthly bond return is potentially useful for dynamic asset allocation. Predicting deviations from that common return is useful for choosing among bonds. A conventional return model combines the two tasks. The decomposition below separates them. It shows that a high out-of-sample R^2 need not imply a proportionally high rank IC, and that an improvement in R^2 need not translate into an improvement in long-short selection.

3 Framework

This section formalizes why a model trained to forecast returns and a model trained to forecast relative performance can rank bonds differently, even when both use the same information. The argument does not claim that the conditional mean is the wrong object in population; it is not. It shows how pooled estimation combines two sources of covariation between characteristics and returns: variation across months, which forecasts the component of returns common to all bonds, and variation across bonds within a month, which forecasts relative performance. Only the second is useful for selection. The return target mixes the two, whereas relative targets use only the second.

3.1 Setup

Month t contains N_t bonds. Bond i has characteristics $Q_{it} \in \mathbb{R}^K$ observed at the end of month t and excess return $R_{i,t+1}$ over month $t+1$. For any variable Z_{it} , let $\bar{Z}_t = N_t^{-1} \sum_i Z_{it}$ be its cross-sectional mean and $\tilde{Z}_{it} = Z_{it} - \bar{Z}_t$ the deviation from that mean. Applied to returns, this gives the identity

$$R_{i,t+1} = M_{t+1} + A_{i,t+1}, \quad M_{t+1} \equiv \bar{R}_{t+1}, \quad A_{i,t+1} \equiv \tilde{R}_{i,t+1}, \quad (1)$$

which splits each return into a common component M_{t+1} and a relative component $A_{i,t+1}$ with $\sum_i A_{i,t+1} = 0$. The decomposition is an accounting identity and imposes no assumption.

Pooled estimation treats every bond-month as one observation. I therefore state moments under the *pooled distribution* $\mathbb{P}^*$, which draws month t with probability proportional to N_t and then draws a bond uniformly within that month. Moments under $\mathbb{P}^*$ carry an asterisk; Var_t and Cov_t denote cross-sectional moments within month t . $\mathbb{P}^*$ can be read as the empirical distribution of the panel, in which case every statement below is an exact identity for pooled least squares, or as its population counterpart under stationarity. Define the between-month and within-month second-moment matrices of the characteristics,

$$\Sigma_B = \text{Var}^*(\bar{Q}_t), \quad \Sigma_W = \mathbb{E}^*[\text{Var}_t(Q_{it})], \quad (2)$$

and assume that all second moments are finite and Σ_W is positive definite.

A *target* is a variable $Y_{i,t+1}$ constructed from the next-month returns $R_{1,t+1}, \dots, R_{N_t,t+1}$ of the bonds in the month.

Definition 1 (Relative and balanced targets). A target is *relative* if it is unchanged when a common shock κ_{t+1} is added to every return in the month. A relative target is *balanced* if, in addition, its cross-sectional mean $\bar{Y}_{t+1}$ equals the same constant in every month.

The demeaned return $A_{i,t+1}$ (mean zero), the percentile rank (mean one-half), and the indicator $\mathbf{1}\{R_{i,t+1} > Q_{0.5,t+1}\}$ of beating the cross-sectional median (mean one-half when N_t is even and there are no ties at the median) are balanced relative targets. The indicator $\mathbf{1}\{R_{i,t+1} > \bar{R}_{t+1}\}$ of beating the cross-sectional average is relative but not balanced: the share of bonds above the average depends on the skewness of the cross-section. The raw return and the absolute indicator $\mathbf{1}\{R_{i,t+1} > 0\}$ are not relative.

3.2 Selection is blind to the common component

Lemma 1 (Dollar-neutral portfolios). *Let the weights w_{it} be measurable with respect to time- t information and satisfy $\sum_i w_{it} = 0$. Then $\sum_i w_{it} R_{i,t+1} = \sum_i w_{it} A_{i,t+1}$. Consequently, the return of the portfolio, and its distribution conditional on time- t information, do not depend on M_{t+1} .*

The lemma applies to every long-short portfolio in this paper, whether equal- or value-weighted. A squared-error forecast, by contrast, is rewarded for predicting M_{t+1} . The next result measures how much of a forecast's accuracy comes from that source.

Lemma 2 (Decomposition of out-of-sample R^2). *For forecasts $\hat{R}_{it}$, define the out-of-sample R^2 relative to a zero forecast,*

$$R_{OOS}^2 = 1 - \frac{\sum_{i,t} (R_{i,t+1} - \hat{R}_{it})^2}{\sum_{i,t} R_{i,t+1}^2}.$$

Then

$$R_{OOS}^2 = s_{TS} R_{TS}^2 + s_{CS} R_{CS}^2, \quad (3)$$

where

$$R_{TS}^2 = 1 - \frac{\sum_t N_t (M_{t+1} - \bar{R}_t)^2}{\sum_t N_t M_{t+1}^2}, \quad R_{CS}^2 = 1 - \frac{\sum_{i,t} (A_{i,t+1} - \bar{R}_{it})^2}{\sum_{i,t} A_{i,t+1}^2},$$

and $s_{TS} = \sum_t N_t M_{t+1}^2 / \sum_{i,t} R_{i,t+1}^2$ and $s_{CS} = 1 - s_{TS}$ are the shares of the common and relative components in the sum of squared returns.

R_{TS}^2 credits a forecast for predicting the average return of the cross-section, both its level and its variation over time. By Lemma 1, that part of accuracy has no value for a dollar-neutral portfolio. Only R_{CS}^2 can be traded in the cross-section.

Proposition 1 (Invariance characterization). *A target is relative if and only if it depends on returns only through the deviations $(A_{1,t+1}, \dots, A_{N_t,t+1})$.*

Remark 1. These results do not imply that the conditional mean ranks bonds incorrectly. Because $\mathbb{E}_t[M_{t+1}]$ is the same for every bond, ranking by $\mathbb{E}_t[R_{i,t+1}]$ is identical to ranking by $\mathbb{E}_t[A_{i,t+1}]$. If the conditional mean were known, the choice of target would not matter for selection. Any difference between targets must therefore come from estimation. The next result shows where it comes from.

3.3 Where the return target goes wrong: between-month contamination

Consider the pooled linear projection of a target on the characteristics, $L^*(Y_{i,t+1} \mid 1, Q_{it}) = \alpha_Y + b'_Y Q_{it}$, with slope $b_Y = \text{Var}^*(Q_{it})^{-1} \text{Cov}^*(Q_{it}, Y_{i,t+1})$. It is the probability limit of pooled least squares and the first-order approximation to any learner that minimizes pooled squared error. Within a month, the forecast ranks bonds by $b'_Y Q_{it}$, which is equivalent to ranking by $b'_Y \tilde{Q}_{it}$ because $b'_Y \bar{Q}_t$ is common to all bonds.

I evaluate a linear score $b'_Y Q_{it}$ by its pooled within-month correlation with the relative

return,

$$\rho(b) = \frac{\mathbb{E}^*[\text{Cov}_t(b'Q_{it}, A_{i,t+1})]}{\sqrt{\mathbb{E}^*[\text{Var}_t(b'Q_{it})] \mathbb{E}^*[\text{Var}_t(A_{i,t+1})]}}, \quad (4)$$

the population analogue of a pooled information coefficient. It depends only on within-month variation and is invariant to positive rescaling of b . Define the within-month and between-month covariances

$$C_W = \mathbb{E}^*[\text{Cov}_t(Q_{it}, A_{i,t+1})], \quad C_B = \text{Cov}^*(\bar{Q}_t, M_{t+1}), \quad (5)$$

and the within-month projection slope $b^* = \Sigma_W^{-1} C_W$. Assume $C_W \neq 0$.

Lemma 3 (Optimal linear ranking). *$\rho(b)$ attains its maximum over $b \neq 0$ exactly at $b = \lambda b^*$ with $\lambda > 0$. For every $b \neq 0$,*

$$\frac{\rho(b)}{\rho(b^*)} = \cos_W(b, b^*) \equiv \frac{b' \Sigma_W b^*}{\sqrt{b' \Sigma_W b} \sqrt{b^{*'} \Sigma_W b^*}} \in [-1, 1]. \quad (6)$$

I call $\cos_W(b, b^*)$ the *ranking efficiency* of the score $b'Q_{it}$: it equals one when the score ranks bonds as well as any linear score can, and minus one when it ranks them in the reverse of the optimal order.

Proposition 2 (Between-month contamination). *For any target Y ,*

$$b_Y = (\Sigma_B + \Sigma_W)^{-1} \left[\underbrace{\text{Cov}^*(\bar{Q}_t, \bar{Y}_{t+1})}_{\text{between months}} + \underbrace{\mathbb{E}^*[\text{Cov}_t(Q_{it}, Y_{i,t+1})]}_{\text{within months}} \right]. \quad (7)$$

In particular:

- (a) *For a balanced relative target, the between-month term is zero. For a relative target that is not balanced, it is $\text{Cov}^*(\bar{Q}_t, \bar{Y}_{t+1})$, where $\bar{Y}_{t+1}$ depends only on the deviations $(A_{j,t+1})_j$ and is unaffected by the common component.*
- (b) *The raw return and the demeaned return satisfy*

$$b_R = b_A + (\Sigma_B + \Sigma_W)^{-1} C_B, \quad b_A = (\Sigma_B + \Sigma_W)^{-1} \Sigma_W b^*. \quad (8)$$

- (c) *For the absolute indicator $\mathbf{1}\{R_{i,t+1} > 0\}$, the between-month term is $\text{Cov}^*(\bar{Q}_t, \pi_{t+1})$, where $\pi_{t+1} = N_t^{-1} \sum_i \mathbf{1}\{M_{t+1} + A_{i,t+1} > 0\}$ is the fraction of bonds with a positive excess return in month $t+1$. For given deviations, π_{t+1} is nondecreasing in M_{t+1} , so the between-month term of this target, like that of the raw return, is driven by the common component.*

Equation (8) is the central result. The slope of the return model equals the slope of a model of relative performance plus a contamination term, $(\Sigma_B + \Sigma_W)^{-1}C_B$. The contamination is nonzero exactly when $C_B \neq 0$, which requires two conditions to hold together: the cross-sectional average levels of the characteristics vary over time ($\Sigma_B \neq 0$), and those levels forecast the common component. Both conditions hold naturally in corporate bonds, where average spreads and yields move with the credit cycle and forecast aggregate bond returns. A pooled return model therefore partly uses characteristics as market-timing variables, and that use leaks into its ranking of bonds. The two corollaries below make the consequences precise.

Corollary 1 (No contamination without level variation). *If the characteristics have no between-month variation ($\Sigma_B = 0$), as when they are demeaned, standardized, or ranked within each month, then $C_B = 0$ and $b_R = b_A = b^*$. The return target and the demeaned-return target then both attain the optimal linear ranking.*

Corollary 2 (Ranking efficiency of the return target). *Suppose $\Sigma_B = \kappa\Sigma_W$ for some $\kappa \geq 0$. Then:*

- (a) $b_A = b^*/(1 + \kappa)$, so the demeaned-return target has ranking efficiency one.
- (b) $b_R = (b^* + \Sigma_W^{-1}C_B)/(1 + \kappa)$. The return target has ranking efficiency one if and only if $C_B = \lambda C_W$ for some $\lambda > -1$, and ranking efficiency minus one if and only if $C_B = \lambda C_W$ for some $\lambda < -1$. In all other cases with $b_R \neq 0$, its ranking efficiency lies strictly between minus one and one.
- (c) With a single characteristic ($K = 1$), the condition $\Sigma_B = \kappa\Sigma_W$ holds automatically, the demeaned-return target always ranks bonds in the optimal order, and the return target ranks them in the optimal order if $C_B/C_W > -1$ and in the reverse order if $C_B/C_W < -1$.

Corollary 2 gives a sharp condition for when the return target distorts the ranking. With several characteristics, the return target is efficient only in the knife-edge case in which each characteristic's power to forecast the common component (C_B) is proportional to its power to forecast relative performance (C_W). Generically this fails, and the return model overweights characteristics whose levels forecast the market. With a single characteristic, the distortion can reverse the ranking: this happens when the characteristic's level forecasts the common component in the opposite direction to its cross-sectional effect, and more strongly.

Remark 2 (Scope). Proposition 2 concerns linear projections, but the incentive it describes does not depend on linearity. Any learner that minimizes pooled squared error and can

condition on characteristic levels lowers its loss by using those levels to predict M_{t+1} . For a balanced relative target the incentive is absent, because the target's cross-sectional mean never varies. Two qualifications apply. First, when characteristic levels vary and Σ_B is not proportional to Σ_W , even the demeaned-return target is distorted by the factor $(\Sigma_B + \Sigma_W)^{-1}\Sigma_W$; Corollary 1 shows that removing level variation from the inputs eliminates both distortions. Second, the probability targets are nonlinear transformations of $A_{i,t+1}$, so their within-month covariances differ from C_W ; what they share with the demeaned return is the absence of a between-month term driven by M_{t+1} .

3.4 Heterogeneous exposure and the information ratio

The identity (1) gives every bond a unit loading on the common component. In the data, bonds differ in their exposure to common rate and credit movements, and, as I show in Section 8, the return model's long-short portfolio has a bond-market beta near one. The next result shows how the two objectives treat such exposure.

Assumption 1 (Factor structure). Conditional on time- t information, $R_{i,t+1} = \beta_i F_{t+1} + E_{i,t+1}$, where $F_{t+1} \sim \mathcal{N}(\mu, \sigma_F^2)$, $E_{i,t+1} \sim \mathcal{N}(e_i, \sigma_E^2)$ with $\sigma_E > 0$, the $E_{i,t+1}$ are independent across bonds and independent of F_{t+1} , and the cross-section is large enough that $\bar{R}_{t+1} = \bar{\beta} F_{t+1} + \bar{e}$.

Let $d_i = \beta_i - \bar{\beta}$ be bond i 's excess exposure and $a_i = e_i - \bar{e}$ its expected relative return net of exposure.

Proposition 3 (Exposure in mean and probability forecasts). *Under Assumption 1,*

$$\mathbb{E}_t[R_{i,t+1} - \bar{R}_{t+1}] = a_i + d_i \mu, \quad \mathbb{P}_t(R_{i,t+1} > \bar{R}_{t+1}) = \Phi(s_i), \quad s_i = \frac{a_i + d_i \mu}{\sqrt{\sigma_E^2 + d_i^2 \sigma_F^2}}, \quad (9)$$

where Φ is the standard normal distribution function. Moreover:

- (a) The conditional mean is linear and unbounded in exposure, with slope μ .
- (b) The probability score is bounded:

$$|s_i| \leq \sqrt{\left(\frac{a_i}{\sigma_E}\right)^2 + \left(\frac{\mu}{\sigma_F}\right)^2}. \quad (10)$$

For a bond without relative alpha ($a_i = 0$) and $\mu \neq 0$, $|s_i| < |\mu|/\sigma_F$ at every level of exposure: the reward for exposure is capped by the Sharpe ratio of the common factor.

(c) $\partial s_i / \partial d_i = (\mu \sigma_E^2 - a_i d_i \sigma_F^2) / (\sigma_E^2 + d_i^2 \sigma_F^2)^{3/2}$. If $a_i > 0$ and $\mu > 0$, the probability score decreases in exposure for $d_i > \mu \sigma_E^2 / (a_i \sigma_F^2)$.

Proposition 3 connects the information-ratio interpretation of probability forecasts (He et al., 2025) to factor exposure. A ranking by expected returns rewards exposure one for one with the factor premium, so it tilts the long leg toward high-exposure bonds and the short leg toward low-exposure bonds. A ranking by the probability of outperformance rewards exposure by at most the factor’s Sharpe ratio, which is small at a monthly horizon, and penalizes it for bonds that already have positive relative alpha. Probability-sorted portfolios should therefore carry less factor exposure than return-sorted portfolios, with the difference concentrated in the long leg, where the return model buys exposure.

The same logic explains why the absolute target $\mathbf{1}\{R_{i,t+1} > 0\}$ behaves differently from the relative ones. Its benchmark is fixed at zero, so its conditional probability depends on the bond’s total volatility, including the volatility of its exposure to the common factor. When the average excess return is positive, bonds with low total volatility have a high probability of a positive return regardless of their relative prospects. The absolute target therefore tends to rank bonds by safety, which lowers the volatility and the factor exposure of the resulting portfolio without improving its ordering of relative performance.

3.5 Two channels and testable predictions

The analysis implies that a relative probability target differs from a return target in two ways, and that each has a distinct portfolio consequence. The relative threshold removes the common component from the target and with it the between-month contamination; this should improve the ordering of bonds and therefore the rank IC. The probability mapping scales expected outperformance by its risk and caps the reward for exposure to the common factor; this should lower the systematic risk and the volatility of the resulting portfolio at a given mean. The absolute probability target shares only the second property and the relative probability targets share both. A return forecast has neither.

The framework identifies two channels through which the return target can rank bonds worse than a relative target. The *between-month channel* (Proposition 2) operates through variation in the average levels of the characteristics and vanishes when the inputs have no between-month variation (Corollary 1). The *exposure channel* (Proposition 3) operates within the month: even with perfectly standardized inputs, a forecast of expected returns rewards exposure to the common factor in proportion to its premium, whereas a probability forecast scales expected outperformance by risk. The framework yields five predictions, which organize the empirical analysis.

- P1.** *Relative targets rank better than the return target; the absolute probability target does not.* Holding the loss fixed, moving the threshold of a probability target from zero to a cross-sectional benchmark removes the between-month term driven by M_{t+1} (Proposition 2(a) and (c)). Among relative targets, a balanced target should do at least as well as an unbalanced one. Section 5.
- P2.** *The measured accuracy of return forecasts is concentrated in the common component.* A return model’s out-of-sample R^2 can come mainly from R_{TS}^2 , which has no value for a dollar-neutral portfolio, so headline R^2 need not track cross-sectional ranking ability (Lemmas 1 and 2). Section 8.
- P3.** *Removing level variation from the inputs removes the between-month channel but not the exposure channel.* For a learner whose inputs are ranked within each month, the linear contamination term is zero. If the relative target still ranks better, part of its advantage comes from the exposure channel. Section 6.
- P4.** *The advantage is larger where characteristic levels and exposures are closely tied to the common component.* The relative target’s advantage should be largest where C_B is large relative to C_W and where exposures to common rate and credit movements differ most across bonds: in long-duration and investment-grade bonds, and in volatile periods (Corollary 2, Proposition 3). Section 8.
- P5.** *Probability-sorted portfolios carry less factor exposure and have lower volatility than return-sorted portfolios, especially in the long leg, so their advantage appears more in risk-adjusted returns and alphas than in mean spreads* (Proposition 3). Sections 5 and 8.

4 Data and Empirical Design

4.1 Sample

I use the monthly corporate bond panel of Bell et al. (2024). The sample contains 107,159 bond-month observations from July 2002 to August 2020 on 1,213 issuers, identified by their CRSP PERMCO, with one bond per issuer and month and a median of 515 bonds per month. The return variable is the winsorized next-month excess return over the risk-free rate. Table 1 reports summary statistics. The average monthly excess return is 0.23 percent with a standard deviation of 2.33 percent. The average numerical rating is 9.3, where higher values denote worse credit quality, and the share of investment-grade bonds ranges from 62 to 78 percent across years. Panel B shows that the cross-section

grows from about 330 bonds per month in 2002 to about 610 after 2013, and that the cross-sectional dispersion of returns is highest in 2002, 2008, and 2020.

Table 1: Summary statistics

Panel A: Bond-month characteristics

	Mean	SD	P5	P25	Median	P75	P95
Excess return (%/month)	0.226	2.334	−2.730	−0.476	0.185	0.970	3.356
Rating (numerical)	9.305	3.190	4.130	7.000	9.000	11.105	15.000
Spread (%)	2.370	1.822	0.536	1.020	1.743	3.270	6.035
Duration (log)	1.602	0.544	0.660	1.368	1.666	1.927	2.344
Maturity (log)	1.870	0.656	0.766	1.532	1.898	2.254	2.875
Amount outstanding (log)	−0.878	1.239	−2.594	−1.851	−1.018	−0.051	1.294
Bond volatility (log)	−4.114	0.638	−5.173	−4.549	−4.116	−3.671	−3.062
Six-month momentum	0.013	0.051	−0.056	−0.003	0.010	0.029	0.086

Panel B: Sample by year

Year	Bond-months	Bonds/month	Issuers	Mean return	CS s.d.	Share IG
2002	1,973	328.8	342	0.613	3.222	0.724
2003	4,523	376.9	414	0.626	1.994	0.677
2004	4,768	397.3	440	0.251	1.306	0.673
2005	4,745	395.4	444	0.104	1.449	0.659
2006	4,624	385.3	430	0.194	1.033	0.674
2007	4,195	349.6	404	−0.446	1.414	0.710
2008	3,984	332.0	366	−0.492	3.131	0.783
2009	4,317	359.8	419	1.415	2.524	0.776
2010	5,312	442.7	497	0.413	1.495	0.702
2011	6,113	509.4	559	−0.062	1.598	0.645
2012	6,655	554.6	638	0.512	1.399	0.617
2013	7,132	594.3	668	0.319	1.028	0.638
2014	7,438	619.8	679	−0.069	1.077	0.642
2015	7,457	621.4	686	−0.162	1.815	0.653
2016	7,338	611.5	663	0.860	1.768	0.662
2017	7,298	608.2	668	0.377	0.943	0.679
2018	7,394	616.2	660	−0.141	1.097	0.708
2019	7,306	608.8	643	0.282	1.105	0.743
2020	4,587	573.4	593	−0.127	2.785	0.764

The sample contains 107,159 bond-months from July 2002 to August 2020 on 1,213 issuers, with one bond per issuer and month. Higher numerical ratings denote lower credit quality. Duration, maturity, amount outstanding, and volatility are in log units, as in the source data. In Panel B, mean return and the cross-sectional (CS) standard deviation are in percent per month; investment grade (IG) is a numerical rating of 10 or better.

4.2 Targets

Let $R_{i,t+1}$ denote the excess return of bond i over month $t + 1$, $\bar{R}_{t+1}$ the equal-weighted cross-sectional average, and $Q_{0.5,t+1}$ the cross-sectional median. I compare four targets:

$$\text{Return:} \quad R_{i,t+1}, \quad (11)$$

$$\text{Prob>Rf:} \quad \mathbf{1}\{R_{i,t+1} > 0\}, \quad (12)$$

$$\text{Prob>Mkt:} \quad \mathbf{1}\{R_{i,t+1} > \bar{R}_{t+1}\}, \quad (13)$$

$$\text{Prob>Median:} \quad \mathbf{1}\{R_{i,t+1} > Q_{0.5,t+1}\}. \quad (14)$$

The return target is estimated under squared error and the three event targets under binary cross-entropy. Because $R_{i,t+1}$ is an excess return, Prob>Rf is the event that the bond beats the risk-free rate. In the terminology of Definition 1, Return and Prob>Rf are absolute, Prob>Mkt is relative but not balanced, and Prob>Median is relative and balanced. The realized cross-sectional statistics enter only the construction of the targets, never the predictors, so they introduce no look-ahead. The unconditional base rates are 0.590 for Prob>Rf, 0.478 for Prob>Mkt, and 0.500 for Prob>Median. The base rate of Prob>Mkt is below one-half because the cross-section of monthly bond returns is right-skewed.

4.3 Predictors

The source data contain 91 candidate predictors after a leakage filter that removes all contemporaneous and future return variables and portfolio weights. I split them into 51 bond- and firm-level characteristics, which vary across bonds within a month, and 40 aggregate variables, which are constant within at least 95 percent of months. The firm-level set includes bond characteristics (rating, spread, yield, duration, maturity, coupon, age, amount outstanding, liquidity), past bond returns and momentum, bond volatility and higher moments, the spread relative to rating and to distance to default, equity characteristics (size, book-to-market, equity momentum and volatility), and accounting ratios (leverage, profitability, debt to EBITDA). Appendix Table 20 lists all predictors.

All models use the 51 firm-level characteristics only. Aggregate variables cannot rank bonds within a month by themselves; they can only help a model forecast the common component of returns or condition firm-level effects on the aggregate state. Excluding them keeps the information set identical across targets and focuses the comparison on cross-sectional information. Missing characteristics are not imputed for the tree-based learners, which treat missingness natively.

4.4 Loss functions

Let $f(X_{i,t}; \theta)$ denote the output of a learner with parameters θ , and let $\mathcal{T}$ be the set of bond-months in the training sample. The return target is estimated by minimizing the pooled mean squared error,

$$\mathcal{L}_{\text{MSE}}(\theta) = \frac{1}{|\mathcal{T}|} \sum_{(i,t) \in \mathcal{T}} (R_{i,t+1} - f(X_{i,t}; \theta))^2. \quad (15)$$

Its population minimizer is the conditional mean $\mathbb{E}[R_{i,t+1} \mid X_{i,t}]$. For an event target $Y_{i,t+1} \in \{0, 1\}$, the learner outputs a log-odds score f , the predicted probability is $p_{i,t} = \Lambda(f(X_{i,t}; \theta))$ with $\Lambda(z) = 1/(1 + e^{-z})$, and the parameters minimize the pooled binary cross-entropy,

$$\mathcal{L}_{\text{BCE}}(\theta) = -\frac{1}{|\mathcal{T}|} \sum_{(i,t) \in \mathcal{T}} [Y_{i,t+1} \log p_{i,t} + (1 - Y_{i,t+1}) \log (1 - p_{i,t})]. \quad (16)$$

Its population minimizer is the conditional probability $\mathbb{P}(Y_{i,t+1} = 1 \mid X_{i,t})$. Binary cross-entropy is a strictly proper scoring rule, so the fitted probabilities are calibrated in population, and the ranking of bonds by $p_{i,t}$ is identical to the ranking by the score f because Λ is strictly increasing.

The two losses treat large returns differently. Under (15), the gradient with respect to the fitted value of one observation is proportional to the forecast error, so a bond with an extreme realized return exerts an influence on the fit that grows with the size of the return. Under (16), the gradient with respect to the score is $p_{i,t} - Y_{i,t+1}$, which is bounded by one in absolute value: every observation contributes the same bounded amount of information regardless of how far its return lies from the threshold. The three event targets share (16) and therefore share this bounded-influence property. They differ only in the threshold that defines the event, which is what allows the target ladder of Section 5 to separate the effect of the loss from the effect of the threshold.

In all learners, early stopping monitors the same loss on the validation sample that is minimized on the training sample: mean squared error (or its square root) for the return target and binary cross-entropy for the event targets. No learner uses sample weights, so every bond-month enters each loss with equal weight.

4.5 Learners

Explainable boosting machines. The main learner is the EBM (Lou et al., 2013; Nori et al., 2019). An EBM is a generalized additive model with a small number of pairwise

interactions,

$$f(X_{i,t}) = \beta_0 + \sum_{k=1}^K g_k(x_{k,i,t}) + \sum_{(k,l) \in \mathcal{I}} g_{kl}(x_{k,i,t}, x_{l,i,t}), \quad (17)$$

where each shape function g_k is a piecewise-constant function of one characteristic and each g_{kl} a piecewise-constant function of a pair. For the return target, f is the forecast itself; for the event targets, f is the log-odds and enters (16) through Λ . The shape functions are estimated by cyclic gradient boosting: in each round, the algorithm cycles through the characteristics one at a time and fits a shallow tree on that characteristic alone to the gradient of the loss, shrunk by the learning rate. Because each tree uses a single characteristic, the ensemble collapses exactly into the additive form (17). After the main effects are fitted, the algorithm selects the interaction pairs $\mathcal{I}$ with the largest gain on the residuals and fits them in the same way.

I bin each characteristic into at most 256 quantile bins for main effects and 32 bins for interactions, allow ten pairwise interactions, require at least 20 observations per leaf, and use a learning rate of 0.03 with at most 5,000 boosting rounds. Missing values receive a separate bin. The model averages eight bags. Each bag is trained on a random 85 percent of the training months, drawn by month so that all bonds in a month stay together, and every bag uses the full validation year to decide when to stop: boosting ends when the validation loss has not improved for 100 rounds. The final shape functions are averages across bags. This protocol makes the stopping rule chronological and identical across targets, and the only difference between the four EBMs is the loss and the label in (15) or (16).

The additive structure is what makes the EBM useful for this paper. Its prediction is an exact sum of one- and two-dimensional terms, so the contribution of every characteristic can be read off without post-hoc approximation, and the same characteristic can be compared across targets. The model is flexible enough to reproduce the nonlinear corporate-bond benchmark of Bell et al. (2024), but restricted enough that differences between targets can be traced to specific characteristics.

XGBoost. The first alternative learner is gradient-boosted trees (Chen and Guestrin, 2016), which allows unrestricted interactions up to the tree depth. I use trees of depth three, a learning rate of 0.02, a minimum child weight of 50, row subsampling of 80 percent and feature subsampling of 50 percent per tree, an ℓ_2 penalty of one on leaf weights, histogram-based split finding, and at most 3,000 trees. Training stops when the validation loss (root mean squared error for the return target and log loss for the event targets) has not improved for 200 trees. The inputs are the raw characteristics, and

missing values are handled by learned default split directions. Like the EBM, XGBoost is invariant to monotone transformations of the inputs, but it does not remove the between-month variation in characteristic levels.

Neural network. The second alternative learner is the NN3 architecture of [Gu et al. \(2020\)](#): a feed-forward network with three hidden layers of 32, 16, and 8 units. Each hidden layer applies a linear map, batch normalization, and a ReLU activation, and the output layer is linear. For the return target the output is the forecast on the original return scale; for the event targets it is the log-odds, trained with the numerically stable form of (16). The objective adds an ℓ_1 penalty of 10^{-5} on all linear-layer weights. I train with the Adam optimizer, a learning rate of 10^{-3} , and mini-batches of 1,024 observations drawn at random across months, for at most 100 epochs, and stop when the validation loss has not improved for five epochs, retaining the parameters with the lowest validation loss. The forecast is the average across an ensemble of five networks with different random initializations.

Following [Gu et al. \(2020\)](#), the network’s inputs are cross-sectional percentile ranks mapped to $[-1, 1]$ within each month, with missing values set to zero, the cross-sectional median. This transformation gives every characteristic the same distribution in every month, so the inputs have no between-month variation. By Corollary 1, it removes the linear contamination channel of the return target, which makes the network a natural test of prediction P3.

Common protocol. All hyperparameters are fixed ex ante and are identical across targets within a learner. The validation sample is used only for early stopping, not to select targets or hyperparameters. Differences between targets within a learner therefore reflect the target alone.

4.6 Sample splits

Expanding window. My main design is an annual expanding window. For each test year y from 2012 to 2020, the validation sample is year $y - 1$ and the training sample contains all earlier months. The models are re-estimated every year, and the out-of-sample forecasts for the 104 months from January 2012 to August 2020 are concatenated.

Fixed split. The second design estimates each model once, with training data from July 2002 to June 2010 (35,657 observations), validation data from July 2010 to June

2011 (5,763), and test data from July 2011 to August 2020 (65,739 observations, 110 months).

Random-year splits. Both chronological designs evaluate the models on essentially the same period. Section 9 therefore also reports 100 random assignments of calendar years to training, validation, and test samples.

4.7 Evaluation

For each forecast I compute the monthly Spearman rank correlation between the forecast and the realized next-month return, the rank IC, and average it over months. Each month I also sort bonds into deciles of the forecast and form the portfolio that buys decile 10 and sells decile 1 (H–L), with equal weights (EW) and with value weights based on each bond’s market value (VW). I split the H–L return into a long leg, decile 10 minus the average bond, and a short leg, the average bond minus decile 1. Sharpe ratios are annualized. All t -statistics use Newey–West standard errors with three lags (Newey and West, 1987). To compare two targets, I test whether the monthly difference in their IC or H–L return has mean zero, which accounts for the strong correlation between the two series in the spirit of Diebold and Mariano (1995).

5 Targets and Cross-Sectional Ranking

This section tests prediction P1 with the EBM. I first compare the four targets on rank IC and portfolio performance, then decompose the difference between the return target and the median target into the contribution of the loss and the contribution of the threshold. I then compare the targets on risk-adjusted performance, examine the stability of the comparison over time, and use conditional portfolio sorts to test whether the probability forecast contains information that the return forecast lacks.

5.1 Main comparison

Table 2 reports out-of-sample performance in the expanding window. The IC is the same in both panels because it does not depend on portfolio weights.

The two relative targets rank bonds substantially better than the return target. The IC of Prob>Median is 0.204 and that of Prob>Mkt is 0.194, against 0.137 for Return. The IC of the relative targets is also more stable across months: its t -statistic is 7.94 for Prob>Median and 4.62 for Return, although both are estimated from the same 104

Table 2: Out-of-sample performance of the four targets (EBM, expanding window)

Target	IC	t	H–L	t	Vol	Sharpe	Long	t	Short	t
<i>Panel A: Equal-weighted deciles</i>										
Return	0.137	4.62	0.612	3.20	1.817	1.17	0.443	3.37	0.168	2.59
Prob>Rf	0.135	7.06	0.544	5.83	0.925	2.04	0.305	4.85	0.239	4.87
Prob>Mkt	0.194	7.43	0.754	5.43	1.241	2.11	0.483	5.34	0.272	4.50
Prob>Median	0.204	7.94	0.727	5.60	1.137	2.21	0.445	5.83	0.282	4.08
<i>Panel B: Value-weighted deciles</i>										
Return	0.137	4.62	0.565	3.21	1.708	1.15	0.427	3.35	0.138	2.45
Prob>Rf	0.135	7.06	0.457	5.08	1.011	1.57	0.252	5.33	0.205	3.40
Prob>Mkt	0.194	7.43	0.652	5.16	1.194	1.89	0.432	5.09	0.221	4.14
Prob>Median	0.204	7.94	0.657	5.92	1.035	2.20	0.433	6.37	0.224	3.71

Annual expanding window; out-of-sample period January 2012 to August 2020 (104 months). IC is the average monthly Spearman correlation between the forecast and the realized excess return. H–L is the return of decile 10 minus decile 1 in percent per month, Vol its monthly standard deviation, and Sharpe its annualized Sharpe ratio. Long is decile 10 minus the average bond and Short the average bond minus decile 1. t -statistics use Newey–West standard errors with three lags. Value weights are bond market values.

months. The long-short portfolios tell the same story in risk-adjusted terms. The equal-weighted Sharpe ratio rises from 1.17 for Return to 2.11 for Prob>Mkt and 2.21 for Prob>Median, and the value-weighted Sharpe ratio from 1.15 to 1.89 and 2.20. The mean H–L return rises less, from 0.61 to 0.73 to 0.75 percent per month with equal weights, and most of the improvement in the Sharpe ratio comes from lower volatility: the monthly volatility of the H–L return is 1.82 percent for Return and 1.14 percent for Prob>Median. Section 5.3 shows that this is the pattern the framework predicts.

The absolute target Prob>Rf behaves differently. Its IC of 0.135 equals that of the return model, but its portfolio has by far the lowest volatility (0.93 percent per month) and therefore a high Sharpe ratio. Its long leg earns only 0.31 percent per month, the smallest of the four. This is the pattern the framework predicts for a target whose benchmark is fixed at zero: the event of a positive excess return is most likely for bonds with low total volatility, so Prob>Rf ranks bonds largely by safety. It lowers the risk of the portfolio without improving its ordering of relative performance.

The fixed split in Appendix Table 21 gives the same ordering. The IC is 0.137 for Return, 0.119 for Prob>Rf, 0.191 for Prob>Mkt, and 0.205 for Prob>Median, and the equal-weighted Sharpe ratios are 1.04, 2.46, 1.51, and 1.93.

5.2 Target ablation: the loss versus the threshold

The four targets form a ladder. Moving from Return to Prob>Rf changes the loss from squared error to cross-entropy and keeps the target absolute. Moving from Prob>Rf

to Prob>Mkt keeps the loss and makes the target relative. Moving from Prob>Mkt to Prob>Median keeps the loss and relativity and makes the target balanced. Table 3 tests each step, together with the direct comparisons against the return target.

Table 3: Paired differences between targets (EBM)

Comparison	Rank IC		H–L EW		H–L VW	
	Δ	t	Δ	t	Δ	t
<i>Panel A: Expanding window</i>						
Prob>Rf – Return (loss only)	−0.002	−0.09	−0.068	−0.40	−0.108	−0.64
Prob>Mkt – Prob>Rf (relative threshold)	0.060	2.91	0.211	1.77	0.195	1.55
Prob>Median – Prob>Mkt (balanced threshold)	0.010	1.61	−0.028	−0.57	0.005	0.09
Prob>Mkt – Return	0.058	3.80	0.142	2.06	0.087	1.35
Prob>Median – Return	0.067	4.99	0.115	1.34	0.092	1.02
Prob>Median – Prob>Rf	0.069	3.47	0.183	1.63	0.199	1.69
<i>Panel B: Fixed split</i>						
Prob>Rf – Return (loss only)	−0.018	−0.89	−0.084	−0.61	−0.130	−0.93
Prob>Mkt – Prob>Rf (relative threshold)	0.072	3.19	0.169	1.58	0.170	1.49
Prob>Median – Prob>Mkt (balanced threshold)	0.014	2.49	0.030	0.88	0.046	1.40
Prob>Mkt – Return	0.054	5.54	0.085	1.55	0.039	0.67
Prob>Median – Return	0.068	6.29	0.115	1.58	0.085	1.13
Prob>Median – Prob>Rf	0.086	4.52	0.199	2.33	0.215	2.40

Each row reports the time-series mean of the monthly difference between two targets’ rank IC or decile H–L return (percent per month) and its Newey–West t -statistic with three lags. The expanding window covers January 2012 to August 2020 (104 months) and the fixed split July 2011 to August 2020 (110 months).

The ladder attributes the gain to the threshold. Changing only the loss has no effect on ranking: the IC difference between Prob>Rf and Return is -0.002 ($t = -0.09$) in the expanding window and -0.018 ($t = -0.89$) in the fixed split. Changing only the threshold, from zero to the cross-sectional average, raises the IC by 0.060 ($t = 2.91$) and 0.072 ($t = 3.19$). The final step to the median adds 0.010 ($t = 1.61$) and 0.014 ($t = 2.49$). The step is small, but its sign matches Proposition 2(a): Prob>Mkt retains a between-month term through the skewness of the cross-section, and Prob>Median does not.

5.3 Mean spread, volatility, and the Sharpe ratio

The ladder shows that the relative threshold improves ordering. The portfolio results in Table 2 show a second difference between the targets, in risk. The mean H–L return of Prob>Median exceeds that of Return by 0.115 percent per month in both designs, with t -statistics of 1.34 and 1.58 , and Prob>Mkt exceeds Return by 0.142 percent ($t = 2.06$) in the expanding window. The difference in mean spreads is therefore modest and not always significant. The difference in risk is large: the monthly volatility of the H–L return

is 1.82 percent for Return and 1.14 percent for Prob>Median in the expanding window, and 2.06 and 1.32 percent in the fixed split.

I emphasize the risk-adjusted comparison for two reasons. First, a long–short portfolio has no natural scale. An investor can lever or delever it, so the relevant measure of a signal’s economic value is the return per unit of risk, not the mean spread of an unscaled decile portfolio. Second, the framework predicts that the relative probability target should improve performance mainly through risk. Proposition 3 shows that a return forecast rewards exposure to the common factor one for one with the factor premium, whereas a probability forecast caps that reward at the factor’s Sharpe ratio. A return-sorted portfolio should therefore carry more systematic risk, which raises both its mean spread and its volatility, while a probability-sorted portfolio should earn a similar spread with less risk (prediction P5).

Table 4: Risk-adjusted comparison with Return (EBM)

Design	Weights	Target	Sharpe ratio				Volatility-matched H–L			
			Target	Return	Δ	t	Target	Return	Δ	t
Expanding	EW	Prob>Rf	2.05	1.17	0.87	2.19	1.524	0.647	0.877	2.21
		Prob>Mkt	2.12	1.17	0.94	5.66	1.378	0.647	0.731	5.66
		Prob>Median	2.22	1.17	1.05	5.62	1.404	0.647	0.757	7.99
	VW	Prob>Rf	1.58	1.15	0.42	1.17	1.296	0.618	0.678	1.59
		Prob>Mkt	1.90	1.15	0.75	4.58	1.228	0.618	0.610	5.45
		Prob>Median	2.21	1.15	1.06	5.01	1.354	0.618	0.737	6.61
Fixed	EW	Prob>Rf	2.48	1.04	1.43	4.58	1.658	0.730	0.927	3.74
		Prob>Mkt	1.52	1.04	0.47	4.79	1.006	0.730	0.276	3.94
		Prob>Median	1.93	1.04	0.89	5.77	1.337	0.730	0.607	8.03
	VW	Prob>Rf	2.20	1.04	1.16	3.88	1.480	0.716	0.764	2.88
		Prob>Mkt	1.47	1.04	0.42	3.41	0.940	0.716	0.224	2.73
		Prob>Median	1.97	1.04	0.92	5.61	1.344	0.716	0.629	7.27

Sharpe ratios are annualized. Δ is the target minus Return. The Sharpe-ratio test follows [Ledoit and Wolf \(2008\)](#): the delta method applied to the first two moments of both return series, with a Newey–West covariance matrix (three lags). The volatility-matched H–L return scales each strategy in month $t + 1$ by a common volatility target, the full-sample monthly volatility of the Return portfolio, divided by the strategy’s rolling standard deviation over the previous twelve months (at least six); the scaling uses only past returns. Means in percent per month; the t -statistic of the difference uses Newey–West standard errors (three lags). The volatility-matched series start once six past months are available, leaving 98 months in the expanding window and 104 in the fixed split.

Table 4 tests the risk-adjusted comparison formally, in two ways. The first is a direct test of the difference in Sharpe ratios. The Sharpe ratio of Prob>Median exceeds that of Return by 1.05 and 1.06 in the expanding window and by 0.89 and 0.92 in the fixed split, with t -statistics between 5.01 and 5.77. The second turns the comparison into a test of mean returns. Each strategy is levered or delevered every month to the same target

volatility, using only its past volatility, so that the two strategies take the same amount of risk ex ante. At equal risk, the Prob>Median strategy earns 1.40 percent per month with equal weights in the expanding window against 0.65 percent for Return, a difference of 0.76 percent ($t = 7.99$). The difference is between 0.61 and 0.76 percent per month in all four design and weighting combinations, with t -statistics between 6.61 and 8.03. The relative target roughly doubles the return per unit of risk, and once risk is equalized, the difference in mean returns that was insignificant for unscaled portfolios becomes large and precisely estimated. Prob>Mkt gives the same result with smaller magnitudes: its volatility-matched advantage is significant in all four cases (t between 2.73 and 5.66).

The absolute target Prob>Rf is the natural control for this comparison, and it shows that low volatility alone is not the result. Its portfolio is the least volatile of the four, and its risk-adjusted advantage over Return is large in the fixed split, where its volatility-matched return exceeds that of Return by 0.93 percent per month ($t = 3.74$). The advantage is not robust, however: in the expanding window with value weights, neither its Sharpe-ratio difference ($t = 1.17$) nor its volatility-matched difference ($t = 1.59$) is significant, and its t -statistics are the lowest of the three targets in every design and weighting scheme except the fixed split with value weights, where they fall between those of Prob>Mkt and Prob>Median. Its IC is no higher than that of Return, it has almost no exposure to the bond-market factor, and its long leg earns the least of the four targets. Prob>Rf ranks bonds largely by safety: when the average excess return is positive, safe bonds are the most likely to have a positive return. It lowers risk without improving the ordering of relative performance, and Section 8 shows that its alpha advantage over Return is not robust across designs. Prob>Median is the only target that combines lower risk with better ordering. Its IC exceeds that of Prob>Rf by 0.069 ($t = 3.47$) and 0.086 ($t = 4.52$), its risk-adjusted advantage over Return is significant with t -statistics above five in every design and weighting scheme, and its alpha advantage over Return is significant in both designs.

The random-year splits in Section 9 confirm this ordering of evidence. Prob>Median has the higher IC in 96 percent of draws, the higher Sharpe ratio in 83 percent, and the higher mean spread in 77 percent. The relative target improves the ordering of bonds and the risk of the portfolio more reliably than it raises the mean spread of an unscaled portfolio, which is the pattern the framework predicts.

The objective-specific losses provide a complementary view. Out of sample, the cross-entropy of Prob>Median is 1.8 percent below that of a constant forecast equal to the base rate, with an area under the ROC curve of 0.593, and the corresponding figures for Prob>Mkt are 1.1 percent and 0.577. The absolute target barely improves on its constant

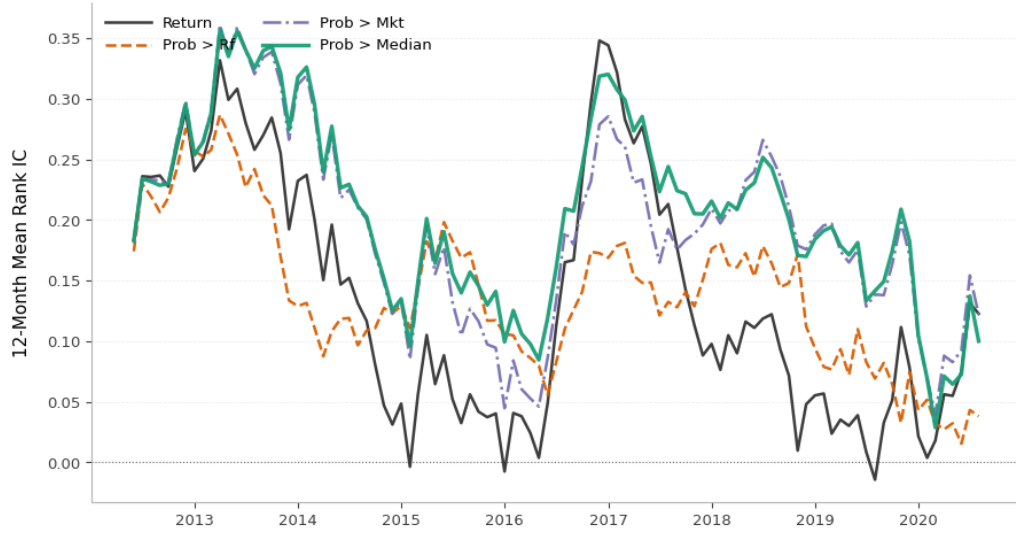
forecast (0.2 percent, AUC 0.541). The return model’s squared error is 0.2 percent above that of a constant forecast equal to the out-of-sample mean (Appendix Table 22). Measured by its own loss, the return model has no predictive skill in the expanding window, although its ranking is informative. In the fixed split the ordering of losses differs: the return model’s squared error is 3.1 percent below the constant forecast, while the cross-entropy skill of Prob>Median is 0.4 percent, with an AUC of 0.589. Statistical fit under each target’s own loss therefore does not line up with ranking performance, which is the same in both designs. This combination anticipates the R^2 decomposition in Section 8: the squared-error loss is dominated by the common component, which the ranking ignores.

5.4 Stability over time

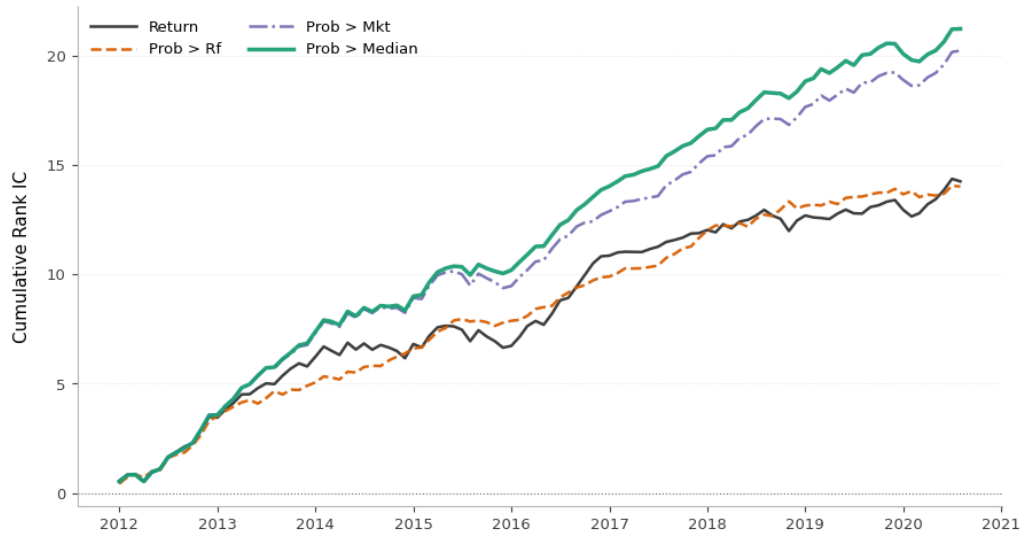
Figure 1 plots the 12-month rolling average and the cumulative sum of the monthly IC in the expanding window. The relative targets separate from the absolute targets early in the sample and the gap widens steadily. By August 2020 the cumulative IC is 21.2 for Prob>Median and 20.2 for Prob>Mkt, against 14.2 for Return and 14.0 for Prob>Rf. The gap opens in two phases. From 2014 to early 2016, a period of falling oil prices and widening high-yield spreads, the cumulative IC of Return is nearly flat while that of Prob>Median keeps rising. From 2017 to 2020, the cumulative IC of Return rises by less than 3 and that of Prob>Median by more than 6. The fixed split in Appendix Figure 8 shows the same pattern, with cumulative ICs of 22.5 for Prob>Median and 15.0 for Return after 110 months.

Figure 2 shows the cumulative returns of the ten decile portfolios for Return and Prob>Median. Both targets produce a monotone ordering of deciles over most of the sample, but the ordering differs in where it is sharp. For the return target, the top decile separates sharply from the rest, while deciles 1 to 8 are bunched together. For Prob>Median, the spread among interior deciles is wider and the bottom decile loses money in cumulative terms, so the long-short return accrues from both legs. The draw-down of March 2020 is visible in both panels, and in both cases it hits the top decile hardest, which is consistent with long positions in bonds with high exposure to the common credit shock. The compounded value-weighted H–L return over the 104 months is 77 percent for Return and 96 percent for Prob>Median; with equal weights, it is 111 percent for Prob>Median.

Appendix Table 23 reports the average return of every decile. Under Prob>Median the average return rises from -0.06 percent per month in decile 1 to 0.67 percent in decile 10, and the increase is spread across the distribution. Under Return it rises from 0.05 to 0.66 percent, with more than half of the increase between deciles 8 and 10.

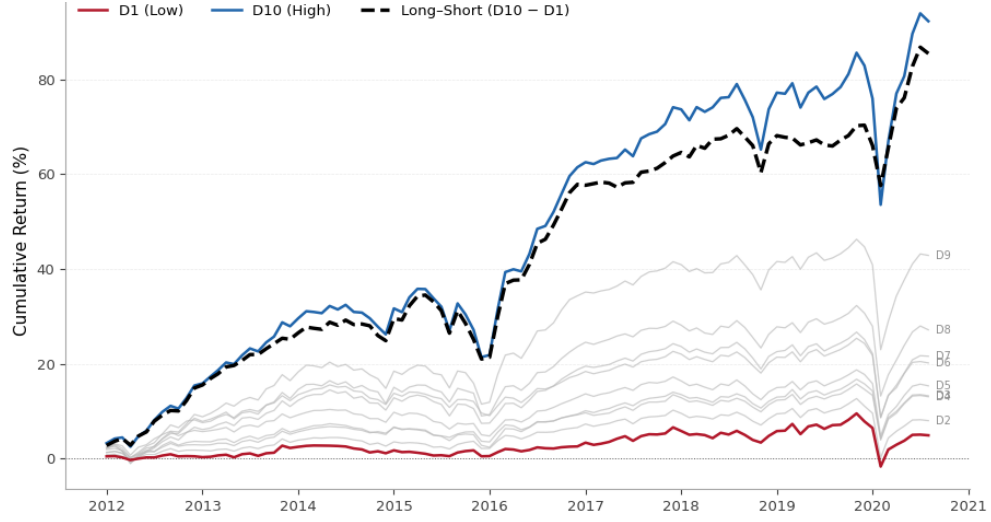


(a) Twelve-month rolling average of the monthly rank IC

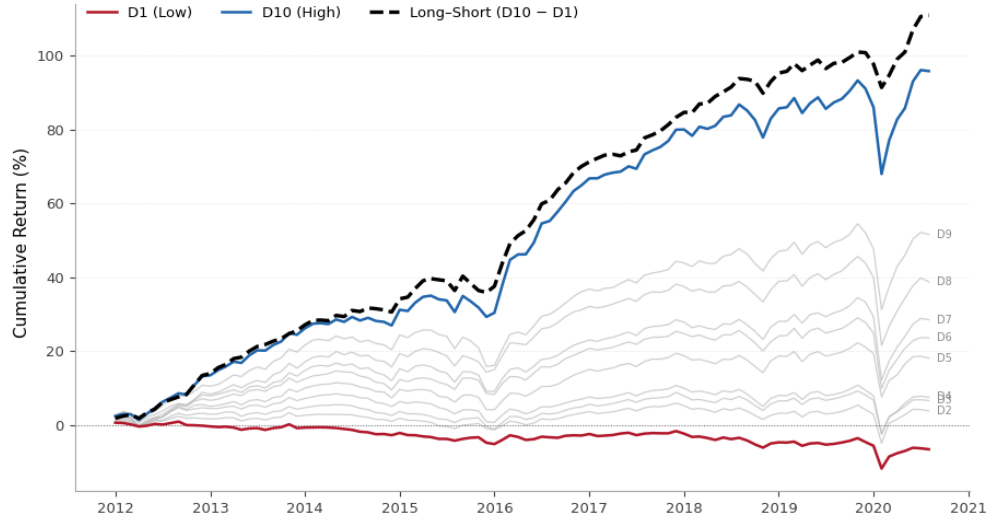


(b) Cumulative rank IC

Figure 1: Rank IC through time (EBM, expanding window). Panel (a) plots the twelve-month rolling average and panel (b) the cumulative sum of the monthly Spearman correlation between each target's forecast and the realized excess return, January 2012 to August 2020.



(a) Return target



(b) Prob>Median target

Figure 2: Cumulative returns of forecast-sorted decile portfolios (EBM, expanding window, equal weights). Each month, bonds are sorted into deciles of the forecast. The figure plots compounded returns of deciles 1 and 10, of the interior deciles (grey), and of the monthly rebalanced decile 10 minus decile 1 portfolio, January 2012 to August 2020.

5.5 Conditional portfolio sorts

The return and probability forecasts share only part of their information: their average monthly rank correlation is 0.59 in the expanding window and 0.55 in the fixed split. A natural question is whether the probability forecast merely repackages the return forecast or contains information about relative performance that the return forecast lacks. If it were a smoother version of the return forecast, sorting on it among bonds with the same return forecast should add little. Conditional double sorts test this directly and without parametric assumptions.

Each month I sort bonds into quintiles of one forecast and, within each quintile, into quintiles of the other. The conditional H–L return of the second forecast is the average across first-sort quintiles of the return difference between its top and bottom quintiles. It measures the second forecast’s ability to rank bonds that the first forecast places in the same group.

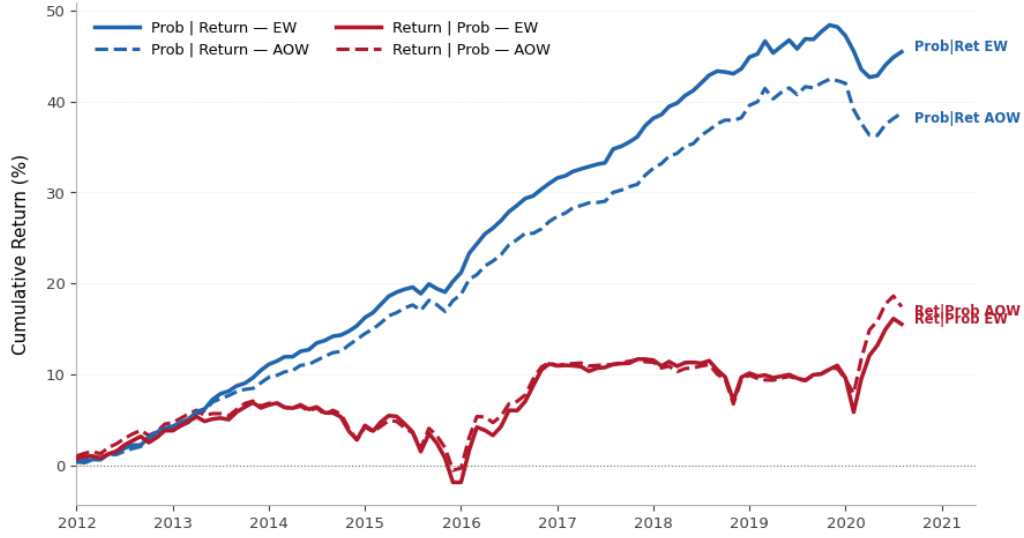
Table 5 reports the results for the EBM in the expanding window. Panel A sorts first on the return forecast and then on Prob>Median. Within every return quintile, the Prob>Median sort produces a positive and significant spread, from 0.23 percent per month in the lowest return quintile to 0.66 percent in the highest. Panel B reverses the order. Within the three lowest Prob>Median quintiles, the return forecast produces spreads between -0.05 and 0.08 percent per month, none of them significant. The return forecast adds information only in the top two Prob>Median quintiles, where its spread reaches 0.47 percent ($t = 2.44$) in the highest quintile.

Panel C averages across first-sort quintiles to give a single conditional long-short strategy for each direction. The contrast is sharpest in risk-adjusted terms. Sorting on Prob>Median within return quintiles earns 0.36 percent per month ($t = 6.02$) in the expanding window with a monthly volatility of only 0.46 percent, an annualized Sharpe ratio of 2.72; in the fixed split it earns 0.39 percent ($t = 8.13$) with a Sharpe ratio of 3.09. Sorting on Return within Prob>Median quintiles earns 0.14 and 0.16 percent ($t = 1.41$ and 1.44) with more than twice the volatility, so its Sharpe ratios are 0.48 and 0.46. With value weights, the Sharpe ratios are between 2.37 and 2.91 for the conditional Prob>Median strategy and between 0.39 and 0.57 for the conditional Return strategy. The conditional Prob>Median strategy thus has a Sharpe ratio four to eight times higher than the conditional Return strategy in every design and weighting scheme. The pattern mirrors the unconditional comparison of Section 5.3: the information the probability forecast adds is not only additional ordering but ordering with little systematic risk. The probability forecast therefore keeps most of its predictive power after conditioning on the return forecast, while the return forecast adds little once the probability forecast is known.

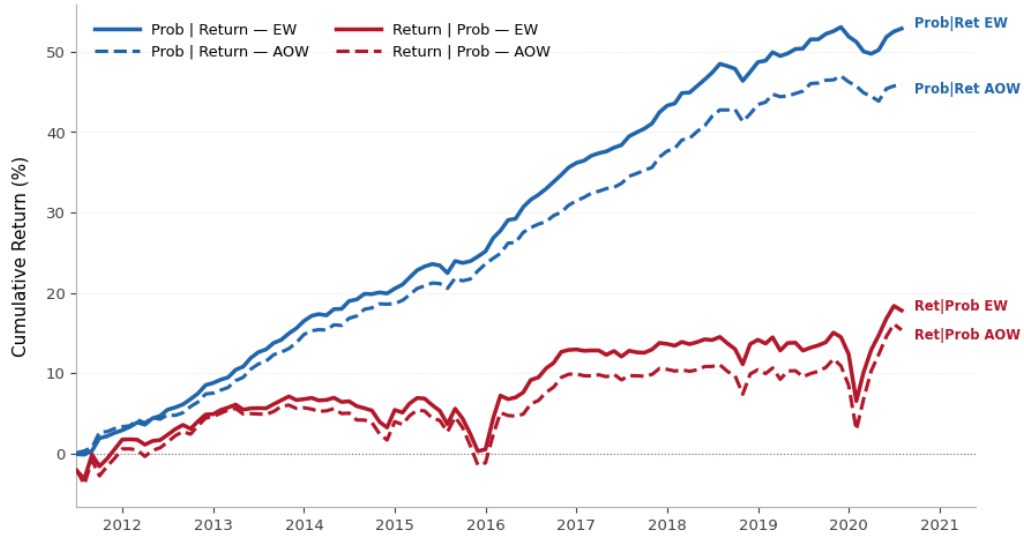
Table 5: Conditional double sorts (EBM)

	Second-sort quintile						
First-sort quintile	1	2	3	4	5	H–L	t
<i>Panel A: First sort on Return, second sort on Prob>Median</i>							
1 (low Return)	−0.072	0.038	0.110	0.097	0.155	0.227	2.91
2	−0.005	0.083	0.106	0.214	0.229	0.234	3.31
3	0.051	0.048	0.126	0.240	0.369	0.318	3.69
4	0.048	0.151	0.234	0.269	0.417	0.369	3.49
5 (high Return)	0.179	0.396	0.498	0.646	0.841	0.662	5.03
<i>Panel B: First sort on Prob>Median, second sort on Return</i>							
1 (low Prob>Median)	−0.033	−0.001	0.008	0.007	−0.025	0.007	0.13
2	0.105	0.065	0.087	0.067	0.050	−0.055	−0.62
3	0.159	0.183	0.195	0.194	0.241	0.082	0.75
4	0.258	0.226	0.243	0.281	0.474	0.217	1.59
5 (high Prob>Median)	0.374	0.357	0.467	0.658	0.844	0.470	2.44
<i>Panel C: Conditional strategies</i>							
Strategy	Mean	t	Vol	Sharpe			
Expanding window							
Prob>Median Return, EW	0.362	6.02	0.461	2.72			
Prob>Median Return, VW	0.316	5.36	0.463	2.37			
Return Prob>Median, EW	0.144	1.41	1.044	0.48			
Return Prob>Median, VW	0.159	1.64	0.963	0.57			
Fixed split							
Prob>Median Return, EW	0.388	8.13	0.435	3.09			
Prob>Median Return, VW	0.345	8.11	0.410	2.91			
Return Prob>Median, EW	0.156	1.44	1.176	0.46			
Return Prob>Median, VW	0.138	1.19	1.231	0.39			

Each month, bonds are sorted into quintiles of the first forecast and, within each quintile, into quintiles of the second forecast. Cells report average next-month excess returns in percent per month. H–L is the return of the top minus the bottom second-sort quintile within the first-sort quintile, with Newey–West t -statistics (three lags). Panels A and B use the expanding window (January 2012 to August 2020) and equal weights. Panel C averages the within-quintile H–L returns across the five first-sort quintiles each month, for both designs and weighting schemes; Mean and Vol are in percent per month and the Sharpe ratio is annualized. Value weights are bond market values.



(a) Expanding window, January 2012 to August 2020



(b) Fixed split, July 2011 to August 2020

Figure 3: Cumulative returns of conditional long-short strategies (EBM). Blue lines sort on the $\text{Prob} > \text{Median}$ forecast within quintiles of the return forecast; red lines sort on the return forecast within quintiles of the $\text{Prob} > \text{Median}$ forecast. Solid lines use equal weights and dashed lines value weights.

Figure 3 plots the cumulative returns of the two EBM conditional strategies in the expanding window. The strategy that sorts on Prob>Median within return quintiles grows steadily, with a cumulative return of about 45 percent with equal weights and 39 percent with value weights. The strategy that sorts on Return within Prob>Median quintiles earns about 16 percent with equal weights and 17 percent with value weights, and most of that arrives after 2016 and in the 2020 recovery. Its path is flat or negative from 2014 through 2016, the same period in which the return model’s cumulative IC stalls in Figure 1. The fixed split in panel (b) shows the same pattern: the conditional Prob>Median strategy reaches a cumulative return of about 53 percent with equal weights, against about 18 percent for the conditional Return strategy.

In sum, the evidence supports prediction P1 and the portfolio part of prediction P5. Relative targets rank bonds better than the return target; the absolute probability target, estimated with the same loss as the relative ones, does not; and the balanced target does slightly better than the unbalanced one. The relative probability target also produces a portfolio that earns a similar mean spread with much less risk, and its forecast contains information about relative performance that the return forecast does not.

6 The Target Effect across Learners

An EBM is an additive model with a few pairwise interactions. If the relative target helps only because it suits this restricted model class, the effect should weaken in more flexible learners. This section re-estimates the Return, Prob>Mkt, and Prob>Median targets with XGBoost and the NN3 network, using the same characteristics, windows, and validation samples. The comparison also tests prediction P3, because the network’s inputs are ranked within each month.

6.1 Performance

Table 6 reports the results for both designs, and Figure 4 summarizes the IC and the equal-weighted Sharpe ratio for all three learners.

The IC ordering of the targets is the same for every learner and design. With XGBoost, the IC is 0.121 for Return, 0.183 for Prob>Mkt, and 0.200 for Prob>Median in the expanding window, and 0.118, 0.187, and 0.204 in the fixed split. With NN3, it is 0.132, 0.215, and 0.215, and 0.138, 0.192, and 0.204. The relative targets reach an IC of about 0.20 regardless of the learner, whereas the return target stays between 0.12 and 0.14. In other words, the level of ranking performance depends much more on the target than on the learner.

Table 6: Out-of-sample performance of XGBoost and NN3

Learner	Target	IC	t	Equal-weighted			Value-weighted		
				H–L	t	Sharpe	H–L	t	Sharpe
<i>Panel A: Expanding window</i>									
XGBoost	Return	0.121	5.08	0.522	3.08	1.22	0.492	3.01	1.18
	Prob>Mkt	0.183	7.03	0.661	4.62	1.73	0.567	4.52	1.66
	Prob>Median	0.200	7.46	0.697	4.79	1.78	0.613	4.66	1.69
NN3	Return	0.132	7.56	0.544	3.95	1.30	0.489	3.55	1.19
	Prob>Mkt	0.215	7.26	0.711	4.75	1.60	0.667	4.52	1.51
	Prob>Median	0.215	7.83	0.696	5.28	1.73	0.631	5.04	1.65
<i>Panel B: Fixed split</i>									
XGBoost	Return	0.118	4.71	0.643	3.80	1.52	0.598	3.71	1.46
	Prob>Mkt	0.187	6.28	0.645	4.19	1.51	0.612	4.56	1.57
	Prob>Median	0.204	7.48	0.691	4.88	1.84	0.634	5.08	1.84
NN3	Return	0.138	7.58	0.652	4.47	1.54	0.583	4.07	1.42
	Prob>Mkt	0.192	6.64	0.642	3.92	1.20	0.587	3.67	1.11
	Prob>Median	0.204	7.44	0.691	4.72	1.44	0.629	4.41	1.34

All learners use the 51 firm-level characteristics and the same training, validation, and test samples as the EBM. XGBoost uses raw characteristics; NN3 uses characteristics ranked within each month and mapped to $[-1, 1]$. H–L in percent per month; Sharpe ratios annualized; Newey–West t -statistics with three lags.

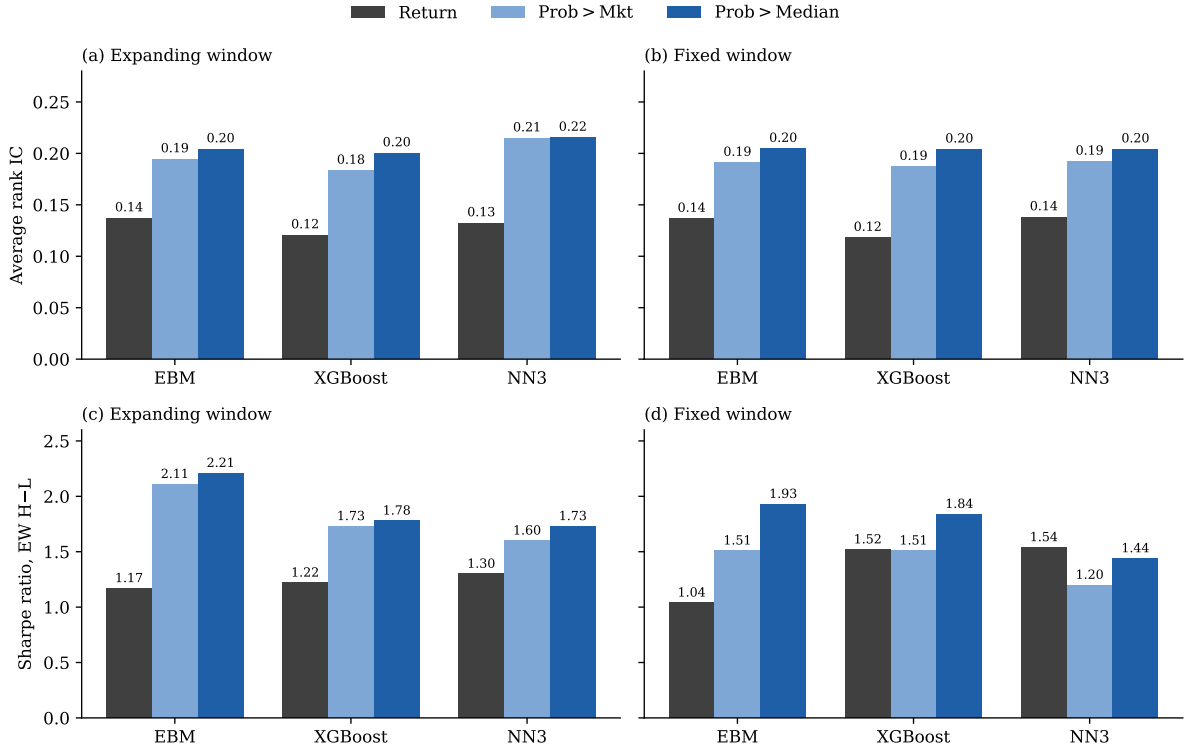


Figure 4: Rank IC and Sharpe ratio by target and learner. Panels (a) and (b) report the average monthly rank IC and panels (c) and (d) the annualized Sharpe ratio of the equal-weighted decile H–L portfolio, for the EBM, XGBoost, and NN3 learners under the expanding window (January 2012 to August 2020) and the fixed split (July 2011 to August 2020).

The Sharpe ratio gains are clear in the expanding window, where Prob>Median raises the equal-weighted Sharpe ratio from 1.22 to 1.78 for XGBoost and from 1.30 to 1.73 for NN3. In the fixed split, Prob>Median improves on Return for XGBoost (1.84 against 1.52) but not for NN3 (1.44 against 1.54), and Prob>Mkt does not improve on Return for either learner. The fixed split trains each model once on data ending in 2010, so its portfolio results depend more on a single fit than those of the expanding window.

6.2 Paired differences

Table 7 tests the within-learner differences.

Table 7: Paired differences between targets within learner

Learner	Comparison	Rank IC		H–L EW		H–L VW	
		Δ	t	Δ	t	Δ	t
<i>Panel A: Expanding window</i>							
XGBoost	Prob>Mkt – Return	0.063	3.68	0.139	1.55	0.075	0.76
	Prob>Median – Return	0.080	4.88	0.175	2.37	0.121	1.33
	Prob>Median – Prob>Mkt	0.017	3.47	0.036	1.01	0.046	1.46
NN3	Prob>Mkt – Return	0.083	4.85	0.167	2.58	0.179	2.57
	Prob>Median – Return	0.083	5.73	0.152	2.40	0.142	1.81
	Prob>Median – Prob>Mkt	0.001	0.17	−0.015	−0.56	−0.036	−1.03
<i>Panel B: Fixed split</i>							
XGBoost	Prob>Mkt – Return	0.069	4.25	0.002	0.02	0.014	0.13
	Prob>Median – Return	0.086	5.95	0.048	0.47	0.036	0.37
	Prob>Median – Prob>Mkt	0.017	2.79	0.046	1.60	0.022	0.81
NN3	Prob>Mkt – Return	0.054	4.05	−0.010	−0.14	0.004	0.05
	Prob>Median – Return	0.066	5.65	0.039	0.52	0.046	0.50
	Prob>Median – Prob>Mkt	0.012	3.60	0.049	1.74	0.042	1.20

Time-series means of monthly differences in rank IC and decile H–L returns (percent per month) with Newey–West t -statistics (three lags).

The IC advantage of Prob>Median over Return is significant in all four learner-design combinations, ranging from 0.066 to 0.086 with t -statistics between 4.88 and 5.95. It is at least as large as for the EBM (0.067 and 0.068). The balanced target Prob>Median improves on Prob>Mkt for XGBoost in both designs ($t = 3.47$ and 2.79) and for NN3 in the fixed split ($t = 3.60$). Differences in mean H–L returns are positive in 15 of the 16 comparisons of Prob>Mkt or Prob>Median against Return and significant in the expanding window for XGBoost with equal weights (0.175, $t = 2.37$) and for NN3 (between 0.142 and 0.179, with t -statistics between 1.81 and 2.58). In the fixed split none of the mean return differences is significant.

6.3 Level variation and the exposure channel

The NN3 results speak directly to prediction P3. Because the network’s inputs are ranked within each month, their cross-sectional distribution is the same in every month and $\Sigma_B = 0$. By Corollary 1, the linear between-month contamination of the return target then vanishes. If contamination through characteristic levels were the only reason the return target ranks worse, the gap between Return and Prob>Median should be small for NN3. It is not: the IC gap is 0.083 in the expanding window and 0.066 in the fixed split, as large as for the learners that see raw characteristics. The return target of NN3 ranks bonds no better than that of the EBM (IC of 0.132 against 0.137).

I draw two conclusions. First, removing level variation from the inputs is not sufficient to align a squared-error model with the ranking task. The between-month channel explains why the absolute and relative probability targets differ in Table 3, but a second channel operates within the month. Second, this within-month channel is the one described by Proposition 3. A forecast of expected returns rewards exposure to the common factor linearly, whether or not the inputs are standardized, whereas a probability forecast scales expected outperformance by its risk. Section 8 provides direct evidence on this channel: return-sorted portfolios carry about twice the bond-market beta of probability-sorted portfolios.

6.4 Agreement across learners

Table 8 reports the average monthly rank correlation between forecasts. Two regularities stand out.

First, learners trained on the relative target agree with one another, whereas learners trained on returns do not. The average rank correlation between the XGBoost and NN3 forecasts is 0.27 for Return and 0.77 for Prob>Median in the expanding window, and 0.21 and 0.78 in the fixed split. The EBM and XGBoost forecasts correlate at 0.68 for Return and 0.88 for Prob>Median. Different flexible learners fit the return target in different ways, which is what one expects when half of the target’s variation is a common component that is unrelated to the ranking and that each learner partly fits in its own way.

Second, within a learner, the return forecast and the probability forecast share only part of their information: their rank correlation is between 0.44 and 0.59. The two relative targets, in contrast, produce almost the same ranking (0.92 to 0.97). The absolute target Prob>Rf correlates with the other targets at 0.41 to 0.56. Its low correlation with both Return and Prob>Median fits the interpretation that it ranks bonds mainly by safety.

Table 8: Agreement of forecasts within and across learners

Pair of forecasts	Expanding	Fixed
<i>Panel A: Same target, different learners</i>		
Return: XGBoost vs. NN3	0.268	0.212
Prob>Mkt: XGBoost vs. NN3	0.760	0.796
Prob>Median: XGBoost vs. NN3	0.767	0.779
Return: EBM vs. XGBoost	0.677	
Prob>Median: EBM vs. XGBoost	0.884	
<i>Panel B: Same learner, different targets</i>		
EBM: Return vs. Prob>Median	0.591	0.552
XGBoost: Return vs. Prob>Median	0.534	0.444
NN3: Return vs. Prob>Median	0.561	0.543
EBM: Prob>Mkt vs. Prob>Median	0.927	0.942
XGBoost: Prob>Mkt vs. Prob>Median	0.927	0.915
NN3: Prob>Mkt vs. Prob>Median	0.971	0.972
EBM: Return vs. Prob>Rf	0.410	0.529
EBM: Prob>Rf vs. Prob>Median	0.560	0.454

Time-series averages of the monthly cross-sectional rank correlation between two forecasts. The EBM versus XGBoost correlations are computed for the expanding window only.

The results of this section support prediction P1 across learners and show that the relative target produces a more stable object. They also qualify prediction P3: ranking the inputs within each month removes one channel of the return target’s disadvantage but leaves the other.

7 Incremental Information across Learners and Regressions

Section 5.5 shows, with conditional portfolio sorts, that the EBM probability forecast contains information that the return forecast lacks, and not the reverse. This section asks whether the asymmetry holds for the other learners and whether it survives a regression that controls for both forecasts at once.

7.1 Conditional sorts across learners

Table 9 summarizes the conditional strategies for all learners, designs, and weighting schemes. For the EBM (Section 5.5), the strategy that sorts on Prob>Median within return quintiles earns between 0.32 and 0.39 percent per month with t -statistics between 5.36 and 8.13 and Sharpe ratios between 2.37 and 3.09. The strategy that sorts on

Return within Prob>Median quintiles earns between 0.14 and 0.16 percent per month, never with a t -statistic above 1.64. XGBoost gives the same asymmetry: the conditional Prob>Median strategy has t -statistics between 5.74 and 7.50, and the conditional Return strategy between 1.62 and 2.07.

Table 9: Conditional long-short strategies across learners

Learner	Design	Weights	Prob>Median Return			Return Prob>Median		
			Mean	t	Sharpe	Mean	t	Sharpe
EBM	Expanding	EW	0.362	6.02	2.72	0.144	1.41	0.48
		VW	0.316	5.36	2.37	0.159	1.64	0.57
	Fixed	EW	0.388	8.13	3.09	0.156	1.44	0.46
		VW	0.345	8.11	2.91	0.138	1.19	0.39
XGBoost	Expanding	EW	0.390	6.11	2.18	0.133	1.67	0.59
		VW	0.365	5.74	2.00	0.126	1.62	0.56
	Fixed	EW	0.405	7.37	2.52	0.172	2.07	0.76
		VW	0.386	7.50	2.31	0.175	2.05	0.76
NN3	Expanding	EW	0.345	4.29	1.44	0.178	3.05	1.19
		VW	0.329	4.43	1.41	0.169	2.84	1.12
	Fixed	EW	0.357	4.72	1.37	0.237	4.04	1.56
		VW	0.354	4.42	1.29	0.211	3.28	1.29

Prob>Median | Return is the average across return-forecast quintiles of the top-minus-bottom Prob>Median quintile return; Return | Prob>Median reverses the roles. Means in percent per month; Sharpe ratios annualized; Newey–West t -statistics (three lags). Value weights are bond market values.

NN3 is the exception in one direction. Its conditional Prob>Median strategy is significant in all four cases (t between 4.29 and 4.72), but its return forecast also retains significant information conditional on Prob>Median (t between 2.84 and 4.04). This is consistent with Section 6. Because the network’s inputs have no between-month variation, its return forecast is not contaminated by the between-month channel, so it carries more relative information that the probability forecast does not already contain. Even so, the conditional Prob>Median strategy earns roughly twice as much as the conditional Return strategy in every NN3 specification.

7.2 Fama–MacBeth horse races

The double sorts condition on one forecast at a time. Fama–MacBeth regressions (Fama and MacBeth, 1973) control for both. Each month I regress the next-month outcome on the cross-sectional percentile ranks of the EBM Return and Prob>Median forecasts, separately and jointly, and average the coefficients over months. I use two outcomes: the excess return, and its cross-sectional percentile rank, which measures the ranking task

directly and is less sensitive to large returns. Table 10 reports the results.

Table 10: Fama–MacBeth regressions on forecast ranks (EBM)

	Expanding window			Fixed split		
	(1)	(2)	(3)	(1)	(2)	(3)
<i>Panel A: Dependent variable is the next-month excess return</i>						
Rank(Return forecast)	0.0051 (2.89)		0.0020 (1.27)	0.0053 (2.91)		0.0018 (1.20)
Rank(Prob>Median forecast)		0.0068 (4.64)	0.0053 (5.13)		0.0072 (4.93)	0.0058 (7.34)
Mean monthly R^2	0.083	0.073	0.106	0.089	0.083	0.117
<i>Panel B: Dependent variable is the rank of the next-month excess return</i>						
Rank(Return forecast)	0.1370 (4.62)		0.0273 (1.07)	0.1365 (4.92)		0.0306 (1.40)
Rank(Prob>Median forecast)		0.2041 (7.94)	0.1866 (8.98)		0.2045 (7.83)	0.1842 (9.36)
Mean monthly R^2	0.098	0.102	0.136	0.094	0.111	0.141

Monthly cross-sectional regressions of the next-month excess return (Panel A, in decimal units) or its cross-sectional percentile rank (Panel B) on the cross-sectional percentile ranks of the EBM forecasts. Coefficients are time-series averages of the monthly estimates, with Newey–West t -statistics (three lags) in parentheses. The expanding window has 104 months and the fixed split 110 months.

In the univariate regressions both forecasts are significant. Because the regressors are percentile ranks, the Panel B coefficients equal the rank ICs of Table 2. In the joint regressions the picture changes. With the excess return as the outcome, the coefficient on the Prob>Median rank falls only slightly, from 0.0068 to 0.0053 in the expanding window, and its t -statistic rises to 5.13; the coefficient on the return rank falls from 0.0051 to 0.0020 and becomes insignificant ($t = 1.27$). A move from the bottom to the top of the Prob>Median ranking is associated with a 0.53 percentage point higher monthly return, holding the return forecast fixed. With the return rank as the outcome, the asymmetry is sharper: the Prob>Median coefficient is 0.187 ($t = 8.98$) and the Return coefficient 0.027 ($t = 1.07$). The fixed split gives nearly identical estimates.

The mean monthly R^2 rises from 0.073 to 0.106 when the return rank is added to the Prob>Median rank in the expanding window, and from 0.083 to 0.106 when the Prob>Median rank is added to the return rank. These increases should be read with care, because a cross-sectional R^2 cannot fall when a regressor is added. The coefficient tests are the informative comparison, and they show that the explanatory power of the return forecast in a joint regression is not reliably different from zero on average.

Taken together, the double sorts and the horse races show that the relative-probability forecast is not a noisier or smoother version of the return forecast. It contains informa-

tion about relative performance that the return forecast lacks, while the return forecast contains little that the probability forecast lacks, except for a learner whose inputs have no between-month variation.

8 Mechanism: Market Timing, Exposure, and Selection

The previous sections establish that relative targets rank bonds better and that their forecasts carry incremental information. This section examines why. I first measure how much of the return models' accuracy comes from the common component (prediction P2). I then ask whether the relative target's ranking reflects bond selection or credit exposure, using characteristic benchmarks, credit-neutral evaluation, and factor regressions (prediction P5). Finally, I locate the effect in the cross-section (prediction P4) and compare what the EBM learns under each target.

8.1 What return models forecast

Table 11 applies the decomposition of Lemma 2 to the return forecasts of the EBM and XGBoost. In the evaluation period, the common component accounts for 47.5 percent of the sum of squared excess returns and the relative component for 52.5 percent.

Table 11: Decomposition of out-of-sample R^2 for return forecasts

Design	Model	R^2_{OOS}	R^2 vs. mean	R^2_{TS}	R^2_{CS}	TS contr.	CS contr.	TS share
Expanding	EBM Return	0.76	-0.22	1.20	0.36	0.57	0.19	75
	XGBoost Return	3.02	2.05	2.83	3.18	1.35	1.67	45
Fixed	EBM Return	3.89	3.11	6.08	1.91	2.89	1.00	74
	XGBoost Return	3.57	2.78	3.92	3.26	1.86	1.71	52

All numbers in percent. R^2_{OOS} is relative to a zero forecast and R^2 vs. mean relative to the out-of-sample mean. R^2_{TS} and R^2_{CS} are defined in Lemma 2. The TS and CS contributions equal $s_{TS}R^2_{TS}$ and $s_{CS}R^2_{CS}$ and sum to R^2_{OOS} ; the TS share is the TS contribution as a percentage of R^2_{OOS} . The variance shares are $s_{TS} = 47.5$ and $s_{CS} = 52.5$ percent in both designs.

For the EBM return model, the time-series component contributes 0.57 of the 0.76 percentage points of R^2_{OOS} in the expanding window and 2.89 of 3.89 in the fixed split, about three quarters in both designs. The cross-sectional R^2 , the only part a long-short investor can use, is 0.36 percent in the expanding window. The fixed split gives a larger headline R^2 of 3.89 percent, but the composition is the same: R^2_{TS} is 6.08 percent and R^2_{CS} 1.91 percent. Because the benchmark is a zero forecast, the time-series part credits a

model both for predicting the average level of bond returns and for timing their variation; neither can be used to rank bonds.

XGBoost allocates its fit differently. The time-series component accounts for 45 percent of its R^2_{OOS} in the expanding window and 52 percent in the fixed split, and its R^2_{CS} is the higher of the two learners, 3.18 against 0.36 percent. Yet XGBoost has the lower IC of the two return models (Table 6). Cross-sectional R^2 and rank IC measure different things: squared error rewards getting the magnitudes of large deviations right, whereas the IC rewards getting the order of all bonds right. A model can fit a few extreme relative returns well and still order the bulk of the cross-section poorly.

The comparison between the two learners illustrates the mechanism of Section 3. A return forecast earns R^2 from two sources, and only one of them is useful for selection. The EBM earns most of its fit from the common component, and XGBoost earns more of it from the relative component, but in neither case does the squared-error fit translate into a ranking comparable to that of the relative target: the IC of both return models is between 0.12 and 0.14, against about 0.20 for Prob>Median (Table 6). Headline R^2 is therefore a poor guide to the cross-sectional value of a return forecast. The evidence supports prediction P2.

8.2 Selection or credit exposure?

If the relative target merely produced a cleaner sort on credit risk, its better ranking would reflect compensated risk rather than selection. I examine this possibility in three ways.

Characteristic benchmarks. Table 12 compares the EBM forecasts with sorts on a single characteristic. A sort on the credit spread has an IC of 0.176 and a Sharpe ratio of 1.10; the EBM return model, with 51 characteristics, has an IC of 0.137 and a Sharpe ratio of 1.17. The return model does not improve on a spread sort in ranking and has similar portfolio volatility. Prob>Median exceeds the spread sort by 0.028 in IC and by 1.1 in Sharpe ratio.

Credit-neutral evaluation. Table 13 evaluates the forecasts within groups of bonds with similar credit quality. Each month I split bonds into rating terciles, spread terciles, or rating-by-duration cells, compute the IC and form portfolios within each group, and average across groups. This removes any contribution from ranking bonds across credit-quality groups.

The return model loses about a third of its IC within rating terciles (from 0.137 to

Table 12: Machine-learning forecasts and characteristic sorts

	IC	t	H–L	t	Vol	Sharpe
<i>Panel A: Expanding window</i>						
EBM Return	0.137	4.62	0.612	3.20	1.817	1.17
EBM Prob>Median	0.204	7.94	0.727	5.60	1.137	2.21
Spread sort	0.176	5.05	0.656	3.24	2.069	1.10
Rating sort	0.126	4.91	0.361	2.70	1.318	0.95
<i>Panel B: Fixed split</i>						
EBM Return	0.137	4.92	0.618	3.10	2.058	1.04
EBM Prob>Median	0.205	7.83	0.733	5.26	1.318	1.93
Spread sort	0.170	4.94	0.647	3.23	2.146	1.04
Rating sort	0.121	4.77	0.353	2.56	1.448	0.85

Equal-weighted deciles. Spread and rating sorts rank bonds by their credit spread and numerical rating at the end of month t . H–L and Vol in percent per month; Sharpe annualized.

Table 13: Credit-neutral evaluation (EBM, expanding window)

Neutral within	Forecast	IC	t	H–L	t	Sharpe
Rating terciles	Return	0.094	4.01	0.379	2.42	0.93
	Prob>Median	0.174	10.28	0.607	6.10	2.64
Spread terciles	Return	0.063	4.86	0.208	3.15	1.09
	Prob>Median	0.148	13.21	0.542	8.61	2.99
Rating $\times$ duration	Return	0.098	4.38	0.296	2.68	1.11
	Prob>Median	0.194	12.31	0.489	6.42	2.78
		Δ IC	t	Δ H–L	t	
Rating terciles	Prob>Median – Return	0.080	4.96	0.228	2.68	
Spread terciles	Prob>Median – Return	0.085	5.81	0.334	3.80	
Rating $\times$ duration	Prob>Median – Return	0.096	6.83	0.194	4.04	

Each month, bonds are assigned to terciles of rating or spread, or to rating-tercile by duration-tercile cells. ICs are computed within groups and averaged; decile portfolios (quintiles for rating $\times$ duration) are formed within groups and pooled with equal weights. H–L in percent per month.

0.094) and more than half within spread terciles (to 0.063). Much of its ranking ability is a sort across credit-quality groups. Prob>Median loses much less (to 0.174 and 0.148), so its advantage grows from 0.067 unconditionally to 0.080 within rating terciles, 0.085 within spread terciles, and 0.096 within rating-by-duration cells, with t -statistics between 4.96 and 6.83. The differences in long-short returns, which were insignificant unconditionally, become significant within credit groups (t between 2.68 and 4.04). Within spread terciles, the Prob>Median portfolio has a Sharpe ratio of 2.99. The fixed split gives the same result: the IC advantage is 0.084 ($t = 6.04$) within rating terciles, 0.095 ($t = 6.69$) within spread terciles, and 0.100 ($t = 7.88$) within rating-by-duration cells, and the long-short differences are significant in all three groupings (t between 2.17 and 4.67).

Composition of the extreme deciles. Table 14 reports the characteristics of the bonds in the extreme deciles. Both targets buy riskier bonds and sell safer ones, but the top decile of Prob>Median is less risky than that of Return, with an average spread of 4.63 against 5.05 percent in the expanding window and 5.07 against 5.35 percent in the fixed split, and lower bond volatility and duration. Its bottom decile contains the highest-rated, lowest-spread, and largest bonds. The return portfolio is closer to a pure long position in credit risk.

Table 14: Characteristics of the extreme deciles (EBM)

Design	Forecast	Decile	Rating	Spread	Duration	Volatility	Amount
Expanding	Return	1	6.71	0.98	1.53	−4.65	−0.70
		10	13.28	5.05	1.61	−3.56	−1.71
	Prob>Median	1	6.12	0.78	1.51	−4.74	−0.45
		10	12.87	4.63	1.46	−3.80	−1.64
Fixed	Return	1	7.32	1.06	1.64	−4.57	−1.00
		10	13.48	5.35	1.61	−3.45	−1.70
	Prob>Median	1	6.11	0.78	1.46	−4.75	−0.49
		10	13.35	5.07	1.47	−3.72	−1.63

Time-series averages of the monthly equal-weighted mean characteristics of bonds in forecast deciles 1 and 10. Spread in percent; duration, volatility, and amount outstanding in log units.

8.3 Factor exposure and alphas

Table 15 regresses the monthly H–L returns of all four EBM targets on the bond-market (MKTB), default (DEF), and term (TERM) factors. Proposition 3 predicts that return-sorted portfolios load more heavily on the common factor than probability-sorted portfolios.

Table 15: Factor regressions of long-short portfolios (EBM)

Target	Weights	α	$t(\alpha)$	β_{MKTB}	β_{DEF}	β_{TERM}	R^2
<i>Panel A: Expanding window (103 months)</i>							
Return	EW	0.483	3.62	1.00	-0.23	-0.54	0.63
	VW	0.454	3.53	0.95	-0.26	-0.52	0.58
Prob>Rf	EW	0.626	5.92	-0.20	-0.02	0.02	0.13
	VW	0.569	5.18	-0.25	-0.13	0.02	0.30
Prob>Mkt	EW	0.688	6.18	0.59	-0.15	-0.36	0.53
	VW	0.582	5.90	0.65	-0.23	-0.37	0.51
Prob>Median	EW	0.671	8.31	0.53	-0.10	-0.33	0.58
	VW	0.613	8.85	0.52	-0.17	-0.32	0.53
<i>Panel B: Fixed split (109 months)</i>							
Return	EW	0.513	4.25	1.12	-0.24	-0.60	0.72
	VW	0.489	4.31	1.12	-0.30	-0.59	0.70
Prob>Rf	EW	0.524	6.01	0.25	-0.12	-0.14	0.16
	VW	0.465	5.20	0.17	-0.14	-0.10	0.05
Prob>Mkt	EW	0.632	6.34	0.82	-0.15	-0.46	0.71
	VW	0.565	5.78	0.76	-0.17	-0.42	0.64
Prob>Median	EW	0.717	7.24	0.55	-0.09	-0.35	0.61
	VW	0.645	7.19	0.59	-0.18	-0.35	0.57

Time-series regressions of monthly decile H–L returns (percent per month) on the bond-market, default, and term factors realized over the same month. α in percent per month, with Newey–West t -statistics (three lags). One month is lost in each design because of the one-month lead applied to the factor series.

The return model’s long-short portfolio has a bond-market beta of 1.00 in the expanding window and 1.12 in the fixed split, even though a long-short portfolio could in principle be market neutral, and the three factors explain 63 to 72 percent of its variance. The Prob>Median portfolio has a beta of 0.53 to 0.59. The absolute target Prob>Rf has almost no market exposure and a factor R^2 of 0.05 to 0.30, which is consistent with its ranking bonds by safety rather than by relative prospects. Prob>Mkt lies in between.

After adjusting for factor exposure, the relative targets deliver substantially more alpha. The Prob>Median portfolio earns 0.67 percent per month ($t = 8.31$) with equal weights and 0.61 percent ($t = 8.85$) with value weights in the expanding window, against 0.48 ($t = 3.62$) and 0.45 percent ($t = 3.53$) for Return. Table 16 tests the differences directly, by regressing the difference between two long-short returns on the factors. The alpha difference between Prob>Median and Return is 0.19 percent per month ($t = 2.55$) with equal weights in the expanding window and 0.20 percent ($t = 5.00$) in the fixed split. This result completes the risk-adjusted comparison of Section 5. The unconditional mean spreads of the two portfolios are close, but a larger part of the return portfolio’s spread is compensation for bond-market exposure. Once that exposure is priced, the relative target earns significantly more, and the difference is estimated precisely because factor adjustment removes the systematic volatility that makes raw mean differences noisy. The absolute target shows why both properties are needed: Prob>Rf earns a high alpha in the expanding window, but its alpha advantage over Return is 0.144 ($t = 1.94$) there and only 0.011 ($t = 0.16$) in the fixed split.

Table 16: Alpha differences and long-leg exposure

		Expanding window				Fixed split			
		$\Delta\alpha$ EW	t	$\Delta\alpha$ VW	t	$\Delta\alpha$ EW	t	$\Delta\alpha$ VW	t
<i>Panel A: Alpha of the difference in H–L returns, target minus Return (EBM)</i>									
Prob>Rf		0.144	1.94	0.114	1.71	0.011	0.16	−0.024	−0.29
Prob>Mkt		0.205	3.97	0.128	2.38	0.119	2.91	0.076	1.75
Prob>Median		0.188	2.55	0.159	1.93	0.204	5.00	0.157	3.52
XGBoost Prob>Median – XGBoost Return		0.170	2.49	0.132	1.50	0.087	1.13	0.069	0.85

Learner	Target	α	$t(\alpha)$	β_{MKTB}	R^2
<i>Panel B: Long leg (decile 10 minus average bond), expanding window, EW</i>					
EBM	Return	0.358	3.62	0.69	0.56
	Prob>Mkt	0.460	5.36	0.25	0.25
	Prob>Median	0.438	6.84	0.17	0.18
XGBoost	Return	0.366	3.65	0.67	0.54
	Prob>Median	0.391	5.04	0.31	0.32

Panel A regresses the monthly difference between two decile H–L returns on the bond-market, default, and term factors and reports the intercept in percent per month. Panel B regresses the long-leg return on the same factors. Newey–West t -statistics with three lags.

Panel B of Table 16 shows that the difference in exposure is concentrated in the long leg, as Proposition 3 predicts. The long leg of the EBM return model has a bond-market beta of 0.69 and a factor R^2 of 0.56. The long leg of Prob>Median has a beta of 0.17 and an R^2 of 0.18, and still earns an alpha of 0.44 percent per month ($t = 6.84$). XGBoost shows the same pattern, with long-leg betas of 0.67 for Return and 0.31 for Prob>Median. The bonds the probability model buys are not simply high-beta bonds. This evidence supports prediction P5.

8.4 Where the target matters

How systematic are bond returns? Table 17 reports the share of excess-return variance explained by monthly means within increasingly fine groups. Month fixed effects alone explain 42 percent of the variance, month-by-rating-tercile means 48 percent, and month-by-rating-by-duration means 53 percent. Within investment-grade and high-yield bonds, month effects explain 45 and 48 percent. The common component is large throughout and larger still once exposures are allowed to differ by rating and duration.

Table 17: Share of return variance explained by common components

Group means	Share of variance
Month	0.423
Month $\times$ IG/HY	0.465
Month $\times$ rating tercile	0.475
Month $\times$ rating tercile $\times$ duration tercile	0.529
Month, investment grade only	0.451
Month, high yield only	0.480
R^2 of monthly excess returns on group-by-month means, full sample July 2002 to August 2020.	

Subsamples. Table 18 compares the two targets within subsamples, with portfolios formed within each subsample. The IC advantage of Prob>Median is concentrated where the return model ranks worst. Among investment-grade bonds, the return model’s IC is 0.096 and its Sharpe ratio 0.58, while Prob>Median reaches an IC of 0.203 (difference 0.106, $t = 5.91$) and a Sharpe ratio of 1.49. Among high-yield bonds the IC difference is 0.023 and insignificant, although the Sharpe ratio of Prob>Median is still twice as high. Among long-duration bonds the IC difference is 0.100 ($t = 5.68$) and the long-short difference 0.31 percent per month ($t = 3.13$); among short-duration bonds the long-short difference is small and insignificant. The fixed split in Panel B reproduces these patterns closely: in investment-grade bonds the IC difference is 0.109 ($t = 6.26$)

and the long-short difference 0.235 ($t = 3.53$), in long-duration bonds 0.091 ($t = 6.83$) and 0.273 ($t = 3.47$), and in high-yield and short-duration bonds the long-short differences are again insignificant. The evidence on market states is weaker. In the expanding window, Prob>Median earns 0.25 percent per month more than Return in months with above-median VIX ($t = 2.27$) and nothing more in calm months. In the fixed split, the high-VIX difference is 0.14 percent ($t = 1.27$). The IC advantage is significant in both states and both designs.

Table 18: The target effect in subsamples (EBM)

Subsample	Return			Prob>Median			Difference			
	IC	H–L	Sharpe	IC	H–L	Sharpe	Δ IC	t	Δ H–L	t
<i>Panel A: Expanding window</i>										
Investment grade	0.096	0.313	0.58	0.203	0.538	1.49	0.106	5.91	0.225	2.22
High yield	0.110	0.620	1.15	0.132	0.792	2.35	0.023	1.13	0.172	0.94
Short duration	0.194	0.688	1.63	0.264	0.742	2.56	0.070	5.09	0.054	1.03
Long duration	0.073	0.415	0.58	0.173	0.721	1.15	0.100	5.68	0.306	3.13
High-VIX months (55)	0.122	0.420	0.77	0.192	0.673	1.98	0.070	3.58	0.253	2.27
Low-VIX months (49)	0.154	0.827	1.67	0.218	0.787	2.47	0.064	3.05	−0.040	−0.33
<i>Panel B: Fixed split</i>										
Investment grade	0.094	0.325	0.58	0.203	0.560	1.33	0.109	6.26	0.235	3.53
High yield	0.123	0.783	1.24	0.138	0.850	2.13	0.015	0.78	0.067	0.38
Short duration	0.186	0.681	1.47	0.262	0.720	2.10	0.076	6.11	0.039	0.71
Long duration	0.080	0.475	0.59	0.171	0.749	1.17	0.091	6.83	0.273	3.47
High-VIX months (58)	0.128	0.462	0.71	0.188	0.599	1.49	0.060	4.50	0.137	1.27
Low-VIX months (52)	0.146	0.792	1.49	0.223	0.882	2.49	0.077	3.83	0.090	0.97

Equal-weighted deciles formed within each subsample. Investment grade is a numerical rating of 10 or better. Short and long duration are the bottom and top duration terciles, formed each month. VIX months are split at the full-sample median of the monthly VIX; the number of months is in parentheses. H–L in percent per month.

These patterns support prediction P4 with one qualification. The advantage is largest in investment-grade and long-duration bonds in both designs: segments in which cross-sectional return differences are dominated by exposure to common rate movements. The evidence that the long-short advantage is larger in volatile months holds in the expanding window but not in the fixed split, so I treat it as suggestive. The month-only variance share is not larger in investment grade (45 percent) than in high yield (48 percent), so the investment-grade result is not explained by a larger common mean. The more precise statement is that the relative target helps most where heterogeneous exposure to common movements, in particular duration exposure to rates, dominates cross-sectional differences. This is where the exposure channel of Proposition 3 predicts the largest distortion of a return forecast.

8.5 What the targets learn

Because the EBM is additive, I can compare directly what each target learns from the same characteristics. Figure 5 reports the normalized importance of the 20 most important main effects of the Prob>Median model under all four targets, and Figure 6 plots the standardized shape functions of the six characteristics with the highest average importance across targets. Both use the fixed-split models.

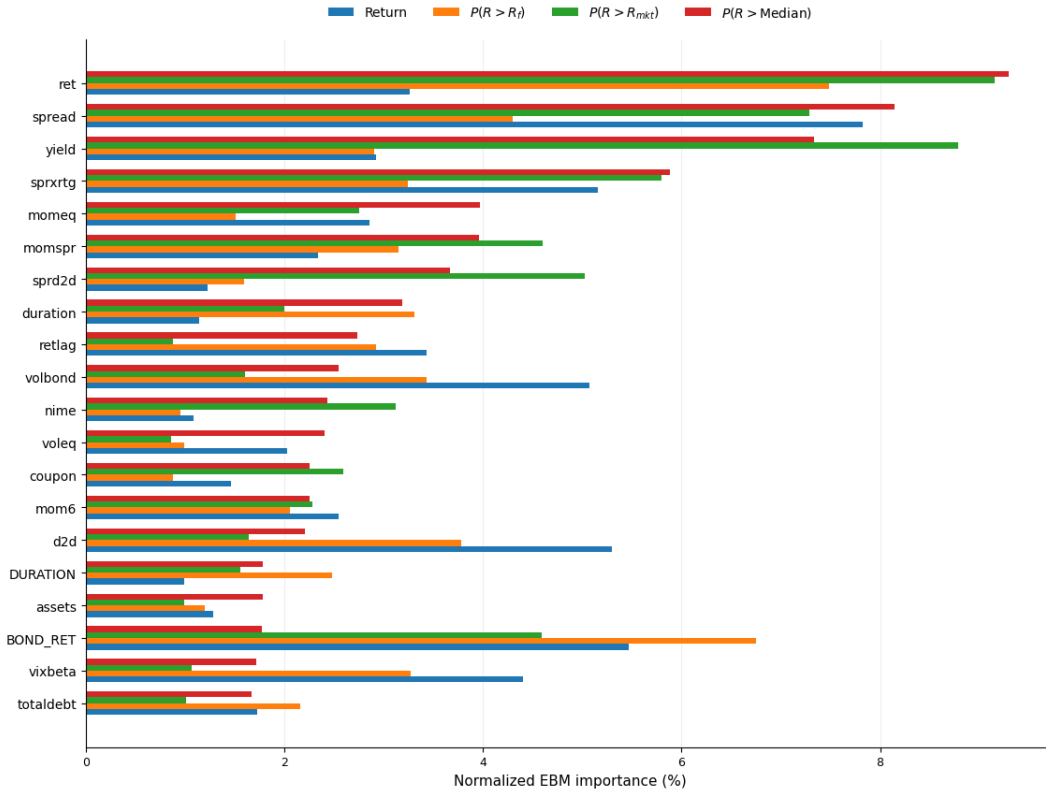


Figure 5: EBM main-effect importance across targets. The figure shows the 20 characteristics with the highest importance in the Prob>Median model and their importance in the Return, Prob>Rf, and Prob>Mkt models. Importance is the mean absolute contribution of a term, normalized to sum to 100 percent within each model. Fixed-split models.

The two relative targets rely on bond-level relative-value signals. For Prob>Median, the most important characteristics are the current month's return (9.3 percent of total importance), the credit spread (8.1), the yield (7.3), the spread relative to rating (5.9), equity momentum (4.0), spread momentum (4.0), and the spread relative to distance to default (3.7). The return model relies more on measures of credit risk and systematic exposure: the spread (7.8), an alternative measure of the bond's current-month return, BOND_RET (5.5), distance to default (5.3), the spread relative to rating (5.2), bond volatility (5.1), and the bond's VIX beta (4.4). Distance to default and VIX beta receive two to three times as much weight in the return model as in Prob>Median. The absolute target Prob>Rf places the most weight on the current return (7.5) and BOND_RET

(6.8) and gives the yield little weight (2.9).

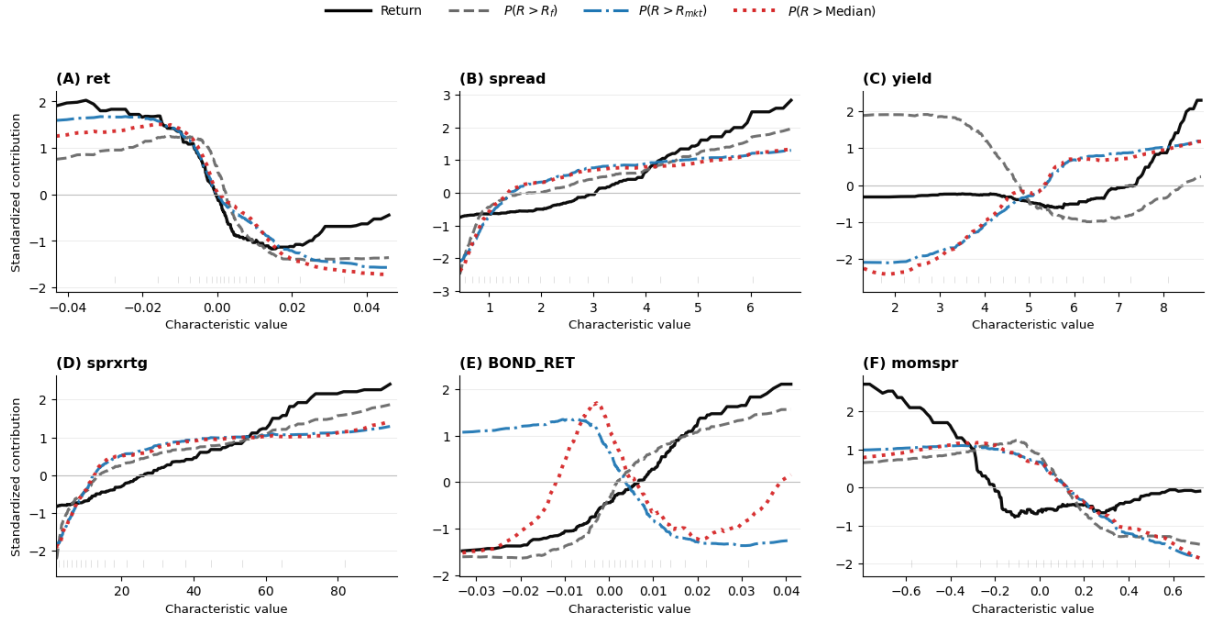


Figure 6: EBM shape functions across targets. Each panel plots the learned contribution of one characteristic to the forecast of each target, standardized to mean zero and unit standard deviation so that targets can be compared on a common scale. The six characteristics have the highest average normalized importance across the four targets. Tick marks show the 5th to 95th percentiles of the characteristic. Fixed-split models.

The shape functions make the difference concrete. For the credit spread and the spread relative to rating, the return model’s contribution rises roughly linearly over the whole range and most steeply for the highest values, where credit premia and return magnitudes are largest. The relative targets’ contributions are concave: they rise steeply among low-spread bonds and flatten in the high-yield range. The relative targets therefore discriminate among investment-grade bonds, where the return model is nearly flat, which matches the subsample results in Table 18. The yield shows the sharpest contrast. The return model’s contribution is flat below a yield of about 6.5 percent and rises steeply above it; the relative targets increase smoothly from low to moderate yields; and Prob>Rf is decreasing over the low range, rewarding low-yield bonds, as expected from a target that ranks by safety. For the current-month return, all targets capture short-term reversal, but the return model’s shape turns up again at large positive returns, where a few extreme observations dominate the squared-error loss, while the relative targets remain monotone.

For BOND_RET, the targets learn different functions. The return model’s contribution increases with it, a momentum pattern, while Prob>Median learns a hump around zero and Prob>Mkt a decreasing, reversal-like shape. In the framework’s terms, recent bond returns are partly a measure of exposure to the common factor, which a forecast of expected returns rewards and a forecast of relative outperformance does not.

9 Robustness

9.1 Evaluation period: random-year splits

The expanding window and the fixed split evaluate the models on essentially the same period, 2011 to 2020. To examine whether the results depend on this period, I draw 100 random partitions of the 19 calendar years from 2002 to 2020 into 11 training years, 3 validation years, and 5 test years, following the random-split designs of [Kozak et al. \(2020\)](#) and [Kaniel et al. \(2023\)](#). Whole years are kept together, training months within six months of any validation or test month are dropped, and in each draw the Return and Prob>Median EBMs are estimated and evaluated on identical samples. Because training years can follow test years, these draws measure the stability of the cross-sectional relation rather than real-time performance. Because the draws share years, the share of draws in which one target wins describes robustness and is not a p -value.

Table 19: Random-year splits: Prob>Median minus Return (EBM)

Test sample	Metric	Mean	5%	Median	95%	Wins (%)
All draws (100)	ΔIC	0.067	0.006	0.063	0.140	96.0
	$\Delta H-L$	0.136	-0.124	0.125	0.410	77.0
	$\Delta Sharpe$	0.656	-0.234	0.492	2.084	83.0
Without 2008 and 2009 (54)	ΔIC	0.086	0.029	0.086	0.165	98.1
	$\Delta H-L$	0.185	-0.072	0.187	0.495	77.8
	$\Delta Sharpe$	0.817	-0.336	0.599	2.410	79.6
With 2008 or 2009 (46)	ΔIC	0.046	-0.015	0.046	0.098	93.5
	$\Delta H-L$	0.078	-0.169	0.111	0.279	76.1
	$\Delta Sharpe$	0.468	-0.103	0.415	1.238	87.0

Each draw assigns calendar years at random to training (11 years), validation (3), and test (5) samples, with a six-month purge of training months around held-out months. The table reports the distribution across draws of the paired difference between Prob>Median and Return in average rank IC, equal-weighted decile H-L return (percent per month), and annualized Sharpe ratio. Wins is the percentage of draws with a positive difference. The lower panels split draws by whether the test years include 2008 or 2009.

Table 19 and Figure 7 summarize the paired differences. The IC result is the most stable: the mean difference is 0.067, the 5th percentile is positive at 0.006, and Prob>Median has the higher IC in 96 of the 100 draws. The Sharpe ratio is higher in 83 draws, with a mean difference of 0.66, and the mean long-short return in 77 draws, with a mean difference of 0.14 percent per month. The ordering, with the IC most stable, then the Sharpe ratio, then the mean return, matches the chronological tests, in which the IC difference is highly significant and the mean-return difference is not. The relative target improves the ordering of bonds more reliably than it raises the average return of the extreme portfolios.

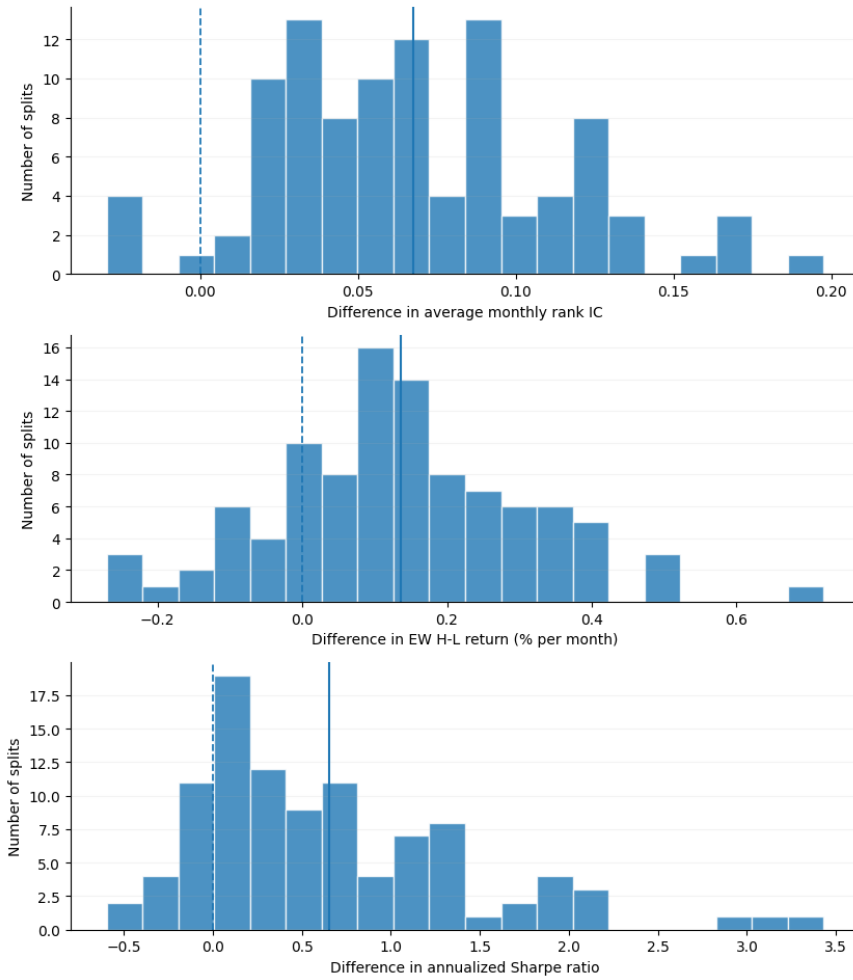


Figure 7: Paired differences between Prob>Median and Return across 100 random-year splits (EBM). The panels show the distribution of the difference in average monthly rank IC, in the mean equal-weighted decile H–L return (percent per month), and in the annualized Sharpe ratio. The solid line marks the mean and the dashed line zero. Positive values favor Prob>Median.

When the test sample includes 2008 or 2009, the IC advantage is smaller (mean 0.046 against 0.086) but still positive in 94 percent of draws. The crisis reduces the size of the effect but does not reverse it. The design cannot separate two explanations. In a crisis, cross-sectional return differences are driven mainly by exposure to the common shock, which favors a model that sorts on exposure; and a draw that tests on the crisis cannot train on it. The chronological evidence is mixed. In the expanding window, the relative target’s advantage in long-short returns is larger in high-VIX months, which favors the second explanation, but the fixed split does not confirm this (Table 18).

9.2 Portfolio weights and sample design

Value-weighted results are close to equal-weighted ones throughout (Tables 2, 3, 6, and 15), so the effect is not confined to small bonds. The fixed split, reported in Appendix Table 21 and throughout the paper, yields IC differences between Prob>Median and Return that are almost identical to those of the expanding window for every learner. The sample has one bond per issuer and month, so the results are not driven by issuers with many bonds, and no model uses sample weights.

9.3 Limitations

Several limitations remain. First, the sample ends in August 2020 and follows the data construction of Bell et al. (2024). Extending it through 2024, including the 2022 rate shock, is a natural out-of-sample test of prediction P4, because that episode was dominated by duration exposure. Second, the Prob>Mkt threshold is the equal-weighted average return of sample bonds rather than a value-weighted index. Third, I report returns before transaction costs. Corporate bond trading costs are substantial, and the return and probability strategies may differ in turnover; since the relative targets’ long legs hold less risky bonds, their costs are unlikely to be higher, but I do not measure them. Fourth, the factor model uses three bond factors from the source data; alphas relative to the factors of Dickerson et al. (2023) would be a stronger test. Fifth, the neural network uses a single set of hyperparameters rather than a validated grid, although the IC ordering of targets is the same for all three learners. Finally, the framework’s formal results concern linear projections and a Gaussian factor structure. The nonlinear learners are consistent with their predictions, but the paper does not establish the mechanism for them formally.

10 Conclusion

Machine-learning research in asset pricing focuses mostly on the choice of learner. My evidence from corporate bonds suggests that the choice of target is at least as consequential. A pooled return model spends much of its measurable accuracy on the component of returns that is common to all bonds in a month, which a long-short investor does not trade, and its cross-sectional ranking is contaminated by the use of characteristic levels to forecast that component. Its ranking is also tilted toward exposure to the common factor, because a forecast of expected returns rewards exposure one for one with the factor premium.

Replacing the return with the probability of beating the cross-sectional median, holding the learner, the predictors, and the samples fixed, raises the rank IC from about 0.13 to about 0.20 for every learner I consider. The resulting portfolio earns a similar mean spread with about half the bond-market beta and much lower volatility, so its Sharpe ratio nearly doubles and its factor-adjusted alpha is significantly higher. This is the risk profile the framework predicts for a probability target, and the ranking reflects selection within credit-quality groups rather than credit exposure. The probability forecast contains information that the return forecast lacks, and not the reverse. The absolute event of a positive return, estimated with the same classification loss, does not rank bonds better than the return forecast. The gain comes from making the target relative, not from the loss.

The broader lesson concerns the alignment of estimation and evaluation. When a forecast will be judged only by its cross-sectional ordering, the target should be invariant to what the ordering ignores. Two tools from this paper apply to any return-prediction study. The decomposition of out-of-sample R^2 shows how much of a reported R^2 can ever be used to select assets, and the ranking-efficiency measure of Lemma 3 shows how much a given target can distort the ranking when characteristic levels forecast the common component.

A Proofs

Throughout, $\mathbb{E}^*$, Var^* , and Cov^* refer to the pooled distribution $\mathbb{P}^*$, under which a month t is drawn with probability proportional to N_t and a bond is then drawn uniformly within that month. Conditional on the month, a pooled moment is a cross-sectional moment: $\mathbb{E}^*[Z_{it} \mid t] = \bar{Z}_t$, $\text{Var}^*(Z_{it} \mid t) = \text{Var}_t(Z_{it})$, and $\text{Cov}^*(Z_{it}, W_{it} \mid t) = \text{Cov}_t(Z_{it}, W_{it})$. Variables dated $t + 1$ are realized quantities of the month drawn, so conditioning on the month also fixes M_{t+1} .

Proof of Lemma 1. By (1), $\sum_i w_{it} R_{i,t+1} = M_{t+1} \sum_i w_{it} + \sum_i w_{it} A_{i,t+1} = \sum_i w_{it} A_{i,t+1}$. The weights are known at t and the right-hand side involves only the deviations $A_{i,t+1}$, so neither the portfolio return nor its conditional distribution depends on M_{t+1} . $\square$

Proof of Lemma 2. Within month t , $\sum_i \tilde{R}_{it} = \sum_i A_{i,t+1} = 0$. Writing $R_{i,t+1} - \hat{R}_{it} = (M_{t+1} - \bar{R}_t) + (A_{i,t+1} - \tilde{R}_{it})$ and expanding, the cross term $2(M_{t+1} - \bar{R}_t) \sum_i (A_{i,t+1} - \tilde{R}_{it})$ vanishes, so

$$\sum_i (R_{i,t+1} - \hat{R}_{it})^2 = N_t (M_{t+1} - \bar{R}_t)^2 + \sum_i (A_{i,t+1} - \tilde{R}_{it})^2.$$

The same argument gives $\sum_i R_{i,t+1}^2 = N_t M_{t+1}^2 + \sum_i A_{i,t+1}^2$. Summing over months,

$$\begin{aligned} 1 - R_{OOS}^2 &= \frac{\sum_t N_t (M_{t+1} - \bar{R}_t)^2 + \sum_{i,t} (A_{i,t+1} - \tilde{R}_{it})^2}{\sum_{i,t} R_{i,t+1}^2} \\ &= s_{TS} (1 - R_{TS}^2) + s_{CS} (1 - R_{CS}^2). \end{aligned}$$

Since $s_{TS} + s_{CS} = 1$, this is equivalent to (3). $\square$

Proof of Proposition 1. Suppose Y depends on returns only through the deviations. Adding κ_{t+1} to every return adds κ_{t+1} to $\bar{R}_{t+1}$ and leaves every deviation unchanged, so Y is unchanged. Conversely, suppose Y is unchanged by every common shock. Choosing $\kappa_{t+1} = -M_{t+1}$ maps the return vector to $(A_{1,t+1}, \dots, A_{N_t,t+1})$, so Y evaluated at the returns equals Y evaluated at the deviations; hence Y is a function of the deviations. $\square$

Proof of Lemma 3. Write $\langle x, y \rangle_W = x' \Sigma_W y$ and $\|x\|_W = \langle x, x \rangle_W^{1/2}$, an inner product because Σ_W is positive definite. By linearity of cross-sectional moments,

$$\mathbb{E}^*[\text{Cov}_t(b' Q_{it}, A_{i,t+1})] = b' C_W = \langle b, b^* \rangle_W, \quad \mathbb{E}^*[\text{Var}_t(b' Q_{it})] = \|b\|_W^2.$$

With $\sigma_A^2 = \mathbb{E}^*[\text{Var}_t(A_{i,t+1})]$,

$$\rho(b) = \frac{\langle b, b^* \rangle_W}{\|b\|_W \sigma_A}.$$

By the Cauchy–Schwarz inequality, $\langle b, b^* \rangle_W \leq \|b\|_W \|b^*\|_W$, with equality if and only if $b = \lambda b^*$ for some $\lambda > 0$. Hence $\rho(b) \leq \|b^*\|_W / \sigma_A = \rho(b^*)$, with equality exactly on that ray, and $\rho(b)/\rho(b^*) = \langle b, b^* \rangle_W / (\|b\|_W \|b^*\|_W) = \cos_W(b, b^*)$. Because $C_W \neq 0$, $b^* \neq 0$ and the ratio is well defined. $\square$

Proof of Proposition 2. By the law of total variance and covariance under $\mathbb{P}^*$,

$$\text{Var}^*(Q_{it}) = \text{Var}^*(\bar{Q}_t) + \mathbb{E}^*[\text{Var}_t(Q_{it})] = \Sigma_B + \Sigma_W,$$

$$\text{Cov}^*(Q_{it}, Y_{i,t+1}) = \text{Cov}^*(\bar{Q}_t, \bar{Y}_{t+1}) + \mathbb{E}^*[\text{Cov}_t(Q_{it}, Y_{i,t+1})].$$

The matrix $\Sigma_B + \Sigma_W$ is positive definite because Σ_W is, so $b_Y = (\Sigma_B + \Sigma_W)^{-1} \text{Cov}^*(Q_{it}, Y_{i,t+1})$ gives (7).

(a) If Y is balanced, $\bar{Y}_{t+1}$ is constant and its covariance with $\bar{Q}_t$ is zero. If Y is relative, each $Y_{i,t+1}$ is a function of the deviations by Proposition 1, and so is $\bar{Y}_{t+1}$.

(b) For $Y = R$, $\bar{Y}_{t+1} = M_{t+1}$, so the between-month term is C_B . Within a month, M_{t+1} is a constant, so $\text{Cov}_t(Q_{it}, R_{i,t+1}) = \text{Cov}_t(Q_{it}, A_{i,t+1})$ and the within-month term is C_W . For $Y = A$, $\bar{Y}_{t+1} = 0$ and the within-month term is again C_W . Therefore $b_A = (\Sigma_B + \Sigma_W)^{-1} C_W = (\Sigma_B + \Sigma_W)^{-1} \Sigma_W b^*$ and $b_R = b_A + (\Sigma_B + \Sigma_W)^{-1} C_B$.

(c) For $Y = \mathbf{1}\{R_{i,t+1} > 0\}$, $\bar{Y}_{t+1} = \pi_{t+1}$. Each indicator $\mathbf{1}\{M_{t+1} + A_{i,t+1} > 0\}$ is nondecreasing in M_{t+1} for fixed $A_{i,t+1}$, and so is their average. $\square$

Proof of Corollary 1. If $\Sigma_B = \text{Var}^*(\bar{Q}_t) = 0$, then $\bar{Q}_t$ is constant $\mathbb{P}^*$ -almost surely, so $C_B = \text{Cov}^*(\bar{Q}_t, M_{t+1}) = 0$. Proposition 2(b) then gives $b_R = b_A = \Sigma_W^{-1} \Sigma_W b^* = b^*$, which is optimal by Lemma 3. $\square$

Proof of Corollary 2. (a) With $\Sigma_B = \kappa \Sigma_W$, $(\Sigma_B + \Sigma_W)^{-1} \Sigma_W = I/(1 + \kappa)$, so $b_A = b^*/(1 + \kappa)$, a positive multiple of b^* , and $\cos_W(b_A, b^*) = 1$.

(b) Similarly, $b_R = (b^* + \Sigma_W^{-1} C_B)/(1 + \kappa)$. Since $\cos_W$ is invariant to positive rescaling, $\cos_W(b_R, b^*) = \cos_W(b^* + \Sigma_W^{-1} C_B, b^*)$. By the equality case of the Cauchy–Schwarz inequality, this equals 1 if and only if $b^* + \Sigma_W^{-1} C_B = \lambda' b^*$ with $\lambda' > 0$, and -1 if and only if the same holds with $\lambda' < 0$. Multiplying by Σ_W , the condition is $C_B = (\lambda' - 1) C_W$. Setting $\lambda = \lambda' - 1$ gives efficiency one for $\lambda > -1$ and minus one for $\lambda < -1$ ($\lambda = -1$ gives $b_R = 0$). If C_B is not a multiple of C_W , b_R is not collinear with b^* and $|\cos_W(b_R, b^*)| < 1$.

(c) For $K = 1$, Σ_B and Σ_W are scalars with $\Sigma_W > 0$, so $\Sigma_B = \kappa \Sigma_W$ with $\kappa =$

$\Sigma_B/\Sigma_W \geq 0$, and $C_B = \lambda C_W$ with $\lambda = C_B/C_W$. Parts (a) and (b) apply. With one characteristic, efficiency one means the score ranks bonds in the same order as b^*Q_{it} and efficiency minus one means the reverse order. $\square$

Proof of Proposition 3. Under Assumption 1, $R_{i,t+1} - \bar{R}_{t+1} = d_i F_{t+1} + (E_{i,t+1} - \bar{e})$. Conditional on time- t information, $d_i F_{t+1} \sim \mathcal{N}(d_i \mu, d_i^2 \sigma_F^2)$ and $E_{i,t+1} - \bar{e} \sim \mathcal{N}(a_i, \sigma_E^2)$ are independent, so $R_{i,t+1} - \bar{R}_{t+1} \sim \mathcal{N}(a_i + d_i \mu, \sigma_E^2 + d_i^2 \sigma_F^2)$. This gives the mean in (9), and $\mathbb{P}_t(R_{i,t+1} - \bar{R}_{t+1} > 0) = \Phi(s_i)$ by symmetry of the normal distribution. Part (a) is immediate.

(b) Write $a_i + d_i \mu = (a_i/\sigma_E) \sigma_E + (\mu/\sigma_F) d_i \sigma_F$ and apply the Cauchy–Schwarz inequality in $\mathbb{R}^2$:

$$|a_i + d_i \mu| \leq \sqrt{(a_i/\sigma_E)^2 + (\mu/\sigma_F)^2} \sqrt{\sigma_E^2 + d_i^2 \sigma_F^2},$$

which is (10). If $a_i = 0$ and $\mu \neq 0$, then $|s_i| = |d_i \mu|/\sqrt{\sigma_E^2 + d_i^2 \sigma_F^2} < |d_i \mu|/(|d_i| \sigma_F) = |\mu|/\sigma_F$ for $d_i \neq 0$ because $\sigma_E > 0$, and $s_i = 0$ for $d_i = 0$.

(c) Let $n_i = a_i + d_i \mu$ and $D_i = (\sigma_E^2 + d_i^2 \sigma_F^2)^{1/2}$. Then

$$\frac{\partial s_i}{\partial d_i} = \frac{\mu D_i^2 - n_i d_i \sigma_F^2}{D_i^3} = \frac{\mu \sigma_E^2 + \mu d_i^2 \sigma_F^2 - a_i d_i \sigma_F^2 - \mu d_i^2 \sigma_F^2}{D_i^3} = \frac{\mu \sigma_E^2 - a_i d_i \sigma_F^2}{D_i^3}.$$

With $a_i > 0$ and $\mu > 0$, the numerator is negative exactly when $d_i > \mu \sigma_E^2/(a_i \sigma_F^2)$. $\square$

B Additional Tables and Figures

Table 20: Predictors

Group	Variables
Bond contract and size	offramt, amtout, maturity, coupon, age, BOND_VALUE, BONDPRC
Credit risk and yields	nrtg, spread, yield, BOND_YIELD, CS, CS_6M_DELTA, sprxrtg, d2d, sprd2d
Duration and rate exposure	duration, DURATION
Bond returns and momentum	ret, retlag, BOND_RET, PREV_BOND_RET, mom6, mom, momspr, mom6xrtg
Bond risk and liquidity	volbond, skew, var, vixbeta, dolvol, turnvol
Equity market	me, beme, dureq, momeq, momeqind, voleq, nime
Accounting	atbe, totaldebt, levmtkt, profitbl, assets, sales, ebitdasale, debtebitda, atme, levop, salesat, chgprof
Aggregate variables (not used)	Bond factors (MKTB, MKTBD, DEF, TERM, DRF, CRF, CRY, DUR, HMLB, LTREVB, MOMB, PEADB, STREVB), equity factors (MKT, SMB, HML, RMW, CMA, MOM, VAL), uncertainty (EPU, EPUT, UNC, UNCf, UNCr and their levels), VIX and its level, liquidity (LIQNT, LIQT, AGGLIQ), intermediary capital (CPTL, INTCAP), RF, RF_chg, CPI

The 51 firm-level characteristics vary across bonds within a month and are used by all models. The 40 aggregate variables are constant within at least 95 percent of months. Variable definitions follow [Bell et al. \(2024\)](#).

Table 21: Fixed split: performance of the four targets (EBM)

Target	IC	t	H-L	t	Vol	Sharpe	Long	t	Short	t
<i>Panel A: Equal-weighted deciles</i>										
Return	0.137	4.92	0.618	3.10	2.058	1.04	0.413	2.79	0.205	3.55
Prob>Rf	0.119	7.12	0.534	6.10	0.750	2.46	0.300	4.87	0.234	7.47
Prob>Mkt	0.191	6.52	0.703	4.39	1.611	1.51	0.449	4.67	0.253	3.43
Prob>Median	0.205	7.83	0.733	5.26	1.318	1.93	0.455	5.38	0.277	4.14
<i>Panel B: Value-weighted deciles</i>										
Return	0.137	4.92	0.589	3.23	1.963	1.04	0.409	2.91	0.180	3.55
Prob>Rf	0.119	7.12	0.459	5.50	0.725	2.19	0.258	4.56	0.201	5.70
Prob>Mkt	0.191	6.52	0.629	4.31	1.490	1.46	0.430	4.54	0.199	3.25
Prob>Median	0.205	7.83	0.674	5.52	1.192	1.96	0.442	5.44	0.233	4.30

Training July 2002 to June 2010, validation July 2010 to June 2011, test July 2011 to August 2020 (110 months). Definitions as in [Table 2](#).

Table 22: Out-of-sample losses relative to constant forecasts (EBM)

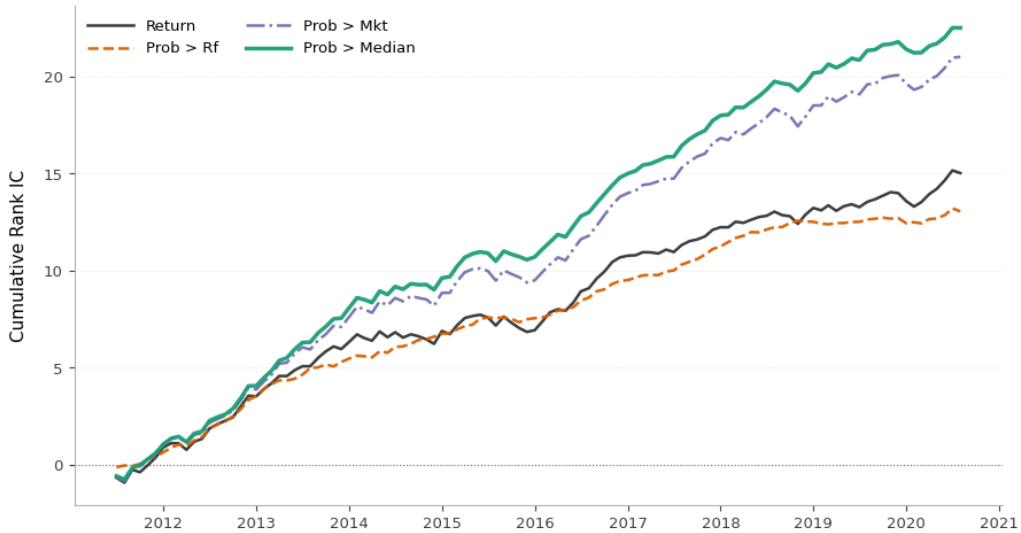
Target (loss)	Expanding window				Fixed split			
	Loss	Constant	Skill	AUC	Loss	Constant	Skill	AUC
Return (MSE)	0.0005	0.0005	−0.002		0.0005	0.0005	0.031	
Prob>Rf (BCE)	0.672	0.673	0.002	0.541	0.667	0.675	0.012	0.582
Prob>Mkt (BCE)	0.685	0.692	0.011	0.577	0.683	0.692	0.013	0.592
Prob>Median (BCE)	0.681	0.693	0.018	0.593	0.690	0.693	0.004	0.589

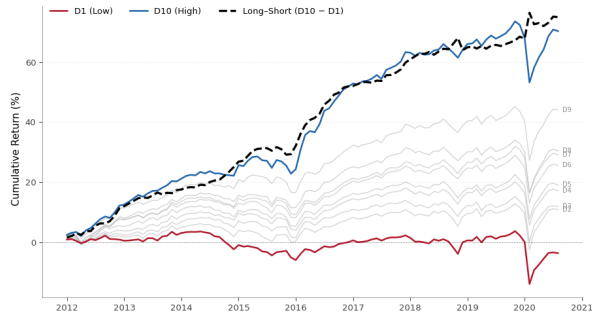
Skill is one minus the ratio of the model’s out-of-sample loss to the loss of a constant forecast equal to the out-of-sample mean or base rate. Squared-error losses are rounded; the skill column uses unrounded values. AUC is the area under the ROC curve.

Table 23: Average excess return by forecast decile (EBM, expanding window)

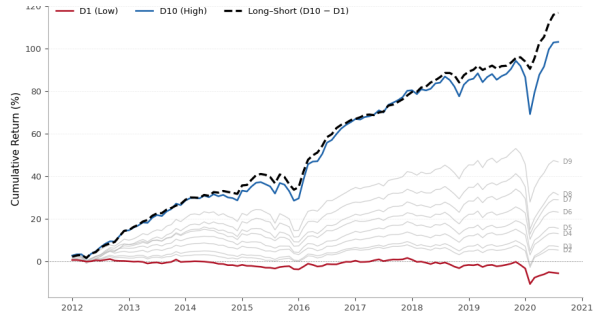
Decile	1	2	3	4	5	6	7	8	9	10
<i>Panel A: Equal weights</i>										
Return	0.052	0.081	0.126	0.126	0.147	0.189	0.202	0.247	0.364	0.664
Prob>Rf	−0.019	0.113	0.121	0.163	0.178	0.230	0.256	0.268	0.363	0.525
Prob>Mkt	−0.051	0.054	0.071	0.125	0.149	0.211	0.261	0.284	0.389	0.703
Prob>Median	−0.061	0.043	0.070	0.079	0.171	0.219	0.260	0.334	0.418	0.665
<i>Panel B: Value weights</i>										
Return	−0.004	0.065	0.098	0.127	0.134	0.168	0.201	0.245	0.362	0.614
Prob>Rf	−0.032	0.122	0.116	0.136	0.170	0.231	0.191	0.247	0.329	0.501
Prob>Mkt	−0.052	0.026	0.048	0.101	0.148	0.146	0.233	0.299	0.409	0.651
Prob>Median	−0.076	0.003	0.037	0.075	0.127	0.158	0.254	0.349	0.425	0.657

Time-series averages of monthly decile returns in percent, January 2012 to August 2020.

**Figure 8:** Cumulative rank IC in the fixed split (EBM). The figure plots the cumulative sum of the monthly Spearman correlation between each target’s forecast and the realized excess return, July 2011 to August 2020.



(a) Prob>Rf



(b) Prob>Mkt

Figure 9: Cumulative returns of forecast-sorted decile portfolios for the Prob>Rf and Prob>Mkt targets (EBM, expanding window, equal weights). See Figure 2 for the Return and Prob>Median targets.

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